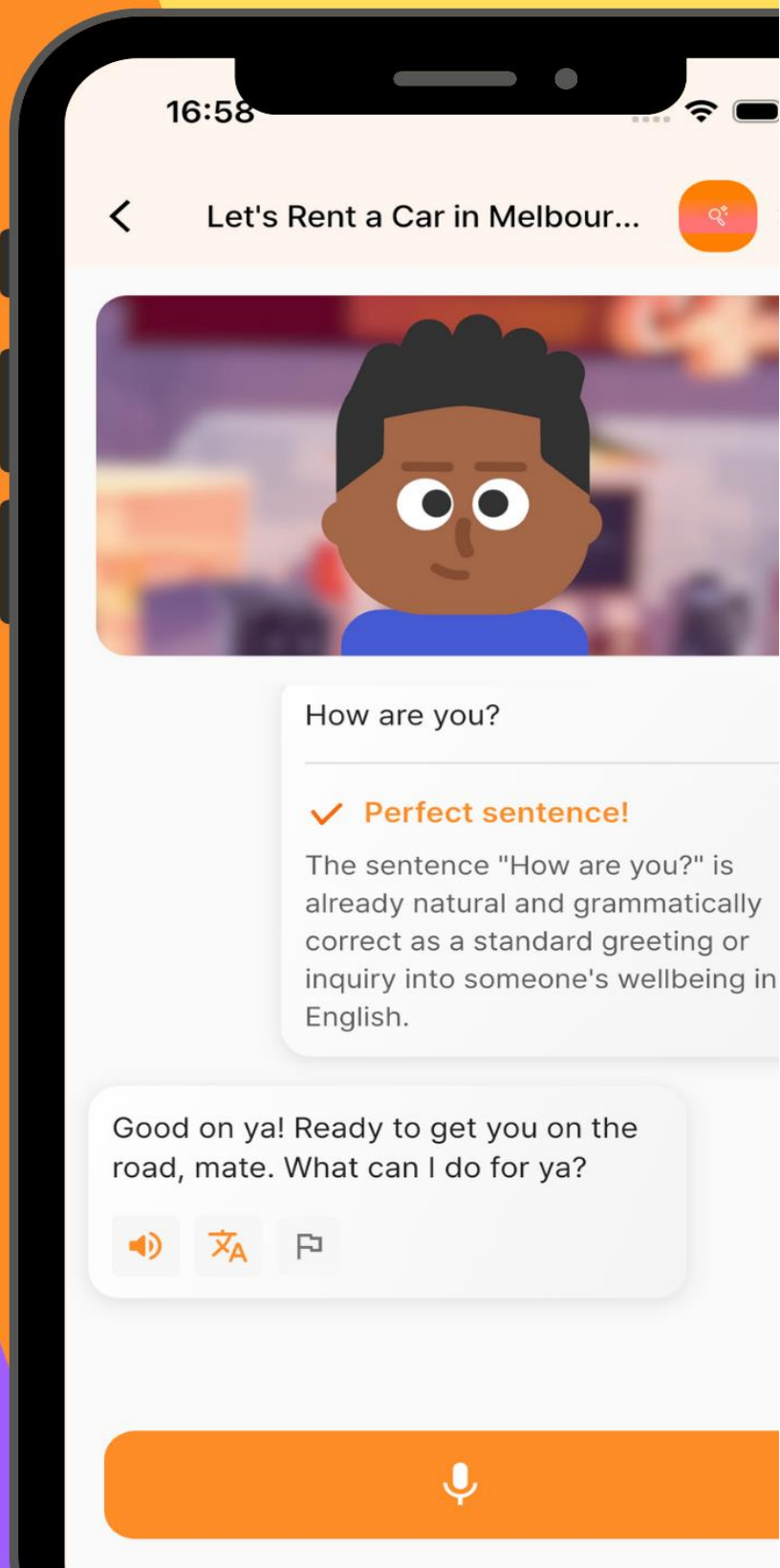
ASM-2: HIGH-FIDELITY DESIGN PROTOTYPE

mojimoji

Natural language learning through AI conversation



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I. Introduction

Link to prototype walkthrough

MojiMoji is a mobile app that provides natural language learning through real-time AI conversations. Users can improve their language skills in realistic conversational environments through role-playing in various situations, receive personalized feedback, and access a convenient progress tracking feature. Experience different conversation styles with three AI characters and enjoy language learning anytime, anywhere with an intuitive interface in 8 languages worldwide.

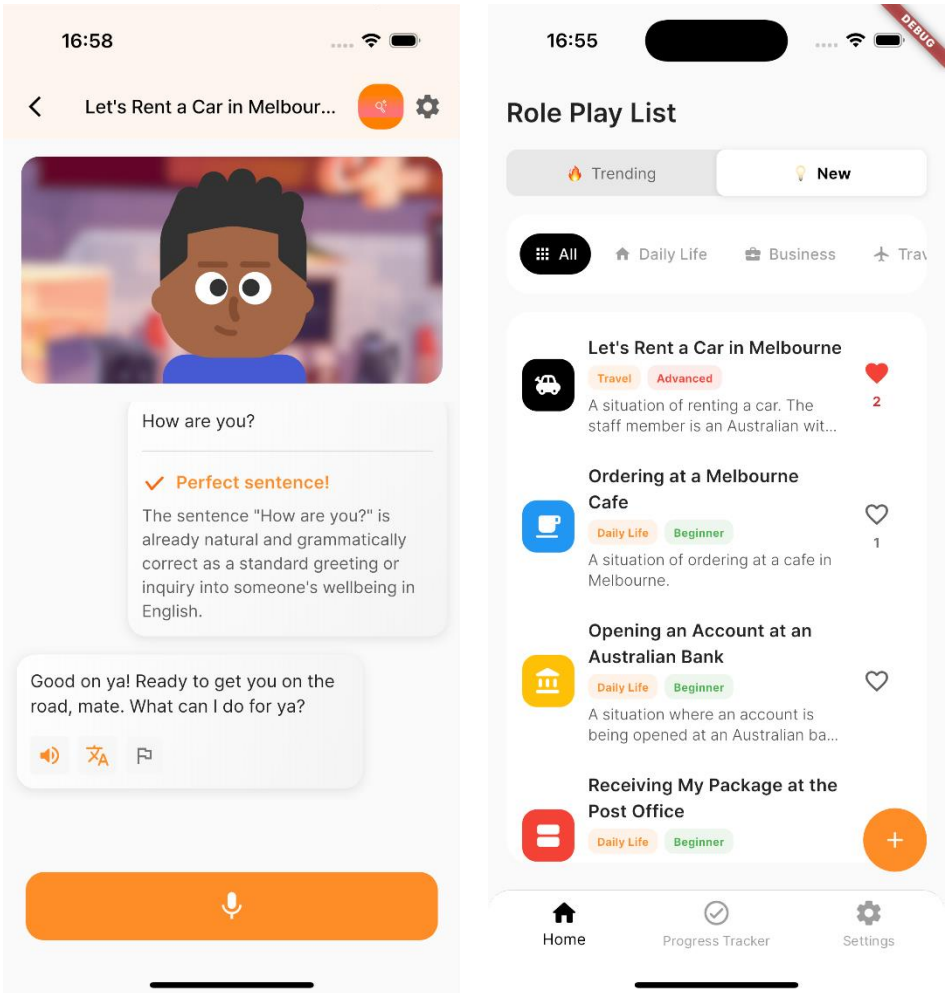


Figure 1: Roleplay scenario screen and home screen

II. System Overview

MojiMoji has the following key objectives

1. **Provide natural conversational language learning:** Improve real communication skills through interaction with AI-based conversation partners.

2. **Offer diverse situational language learning opportunities:** 8 categories of role-play scenarios, including daily life, business, travel, education, entertainment, social, health, and culture.
3. **Personalized learning experience and feedback:** Conversations and detailed feedback tailored to the user's language level and interests.
4. **Multilingual support:** Interface available in 8 languages, including English, Korean, Japanese, Chinese, Hindi, French, and more.
5. **Convenient progress tracking:** Visualization of user activities, scores, and streak days.

MojiMoji has the following core interaction flow

1. **User input:** Users practice language through voice recording.
2. **AI Processing:** Analysis of user input to evaluate grammar, vocabulary, pronunciation, etc.
3. **Real-time feedback:** Natural corrections and alternative expressions suggested within conversations.
4. **Learning data tracking:** Recording of user learning activities, strengths, and areas for improvement.
5. **Progress visualization:** Display of user activities and achievements in graphs and charts.

Input and Output interaction technologies

Input methods:

- Voice recording (during real-time conversations)
- Touch-based menu selection (scenario selection, settings, etc.)

Output methods:

- AI voice responses (various voice types and speaking pace options)
- Visual feedback (grammar corrections, alternative expression suggestions)
- Charts and graphs (learning progress display)
- Animations (feedback on user actions)

III. System Functionalities

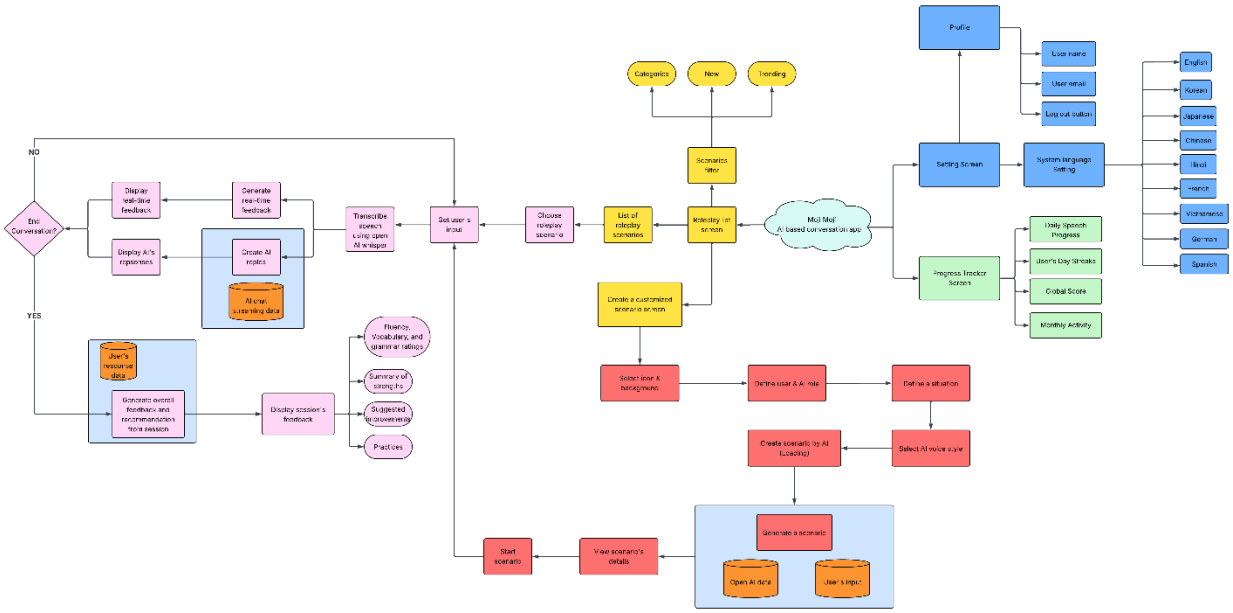


Figure 2: MojiMoji functionalities mind map

[Mindmap link]

Main functionality: The AI roleplay scenario

1. Category selection flow:
    - Categories (daily, business, travel, education, entertainment, social, health, culture)
2. Roleplay list and selection (Main UI components):
    - View role-play scenario (difficulty level, category, descriptions)
3. Roleplay conversation flow:
    - Transcribe user input
    - Generate responses through the OpenAI API
    - Text-to-speech input
    - Grammar and vocabulary analysis and feedback
4. Roleplay customization:
    - Icon selection
    - Character role definition for both the user and the AI
    - Context setting and background establishment
    - AI voice style selection

Progress Tracker:

1. Activity tracking:
    - Calculating day streaks
    - Weekly engagement pattern analysis
2. Global score calculation based on scenario completion:
3. Daily speech tracking
    - Provide detailed analysis on user learning (conversation, word usage, etc.)
    - Word count tracking
    - Time spent learning
4. Monthly activity calendar
    - Provide a visual representation of the user engagement pattern with a heat map style display showing the intensity of usage

Feedback and Recommendation System:

1. Comprehensive performance report
    - Generates detailed analysis, highlighting strengths and areas for improvement
    - Give a rating on fluency, grammar, and vocabulary use

IV. Design choice and development

Design choice rationale

Our design decisions are grounded in established HCI principles (Figma, n.d.)

- **Visibility**: Clear system status through real-time feedback and progress indicators
- **Consistency**: Unified visual language and interaction patterns across all screens
- **Recognition Over Recall**: Interface elements indicate available actions
- **Error Prevention**: Predictive input and confirmation dialogs for potentially destructive actions
- **Minimalist Design**: Focus on essential elements to minimize cognitive load

Accessibility Considerations

The app implements comprehensive accessibility features

- **Motor Accessibility**: Large touch targets (44×44dp minimum), voice commands as input alternatives
- **Cognitive Accessibility**: Clear information architecture, error prevention, consistent navigation patterns

- **Implementation:** Expanded language support from 6 to 8 languages with comprehensive cultural adoption.

Iterative Development of the Prototype

Initial design concept

Our initial low-fidelity prototype focused on creating a language conversation application that facilitates conversation practice through role-play scenarios. The core concept featured a simple structure with a main screen, scenarios, settings, a conversation interface, and feedback components. This minimal viable product approach allowed us to establish the fundamental user flow before adding complexity and refinement.

First Iteration: Scenarios screen enhancement

Changes made:

- Expanded the simple scenario list with more detailed information
- Added descriptions and categories for each scenario
- Implemented visual category icons and difficulty indicators
- Added functionality for users to filter scenarios by categories

Rationale for changes

- The initial low-fidelity prototype only displayed the titles, which did not convey enough information for users to understand the scenario
- Team discussions identified a lack of information needed for users to select scenarios appropriate for their language proficiency level
- Recognizing the need for more contextual information in the scenario selection process to enhance the user experience

Key improvements

- Users can quickly understand the nature and difficulty of each scenario immediately
- Easier to find role plays in areas of interest through category-based filtering
- Provides detailed information to aid user decision-making when selecting the scenario
- Improved information hierarchy leads to increased user satisfaction

Second Iteration: Add Customize Scenario Feature

Changes Made

- Added dedicated functionality for users to create a custom scenario
- Implemented a step-by-step interface for specifying scenario title, situation description, character settings, and difficulty level
- Integrated automatic category system
- Added a toggle option to make the created scenario public or private

Rationale for Changes

- The initial low-fidelity prototype completely lacked the ability for users to custom-make unique scenarios
- High-fidelity prototype aimed to explore the potential for user content contribution and enhance the inclusiveness of the scenarios

Key Improvements

- Users can create customized scenarios tailored to their learning goals and interests
- App content diversity and volume naturally expand through user contributions
- Increased user engagement and time spent on the application
- Enhanced collaborative learning experience through community building

Third Iteration: AI avatar model enhancement

Changes Made

- Added an AI avatar on the selection screen
- Implemented preview of scenarios, expected topic, and learning objectives
- Added difficulty indicator and estimated duration of scenario
- Integrated user review and rating system
- Added 'like' functionality

Rationale for Changes

- The initial low-fidelity prototype had limited pathways for reviewing details before selection
- Team discussions determined that users needed a better understanding of scenario context before starting
- High-fidelity prototype aimed to support more informed user choices

Key Improvements

- Users can clearly understand the content and difficulty level before starting a role play
- Transparent information about expected learning outcomes and time investment

V. Usability Testing of Prototype

Study goal: This testing of our prototype aims to evaluate how effectively, efficiently, and satisfactorily real users can interact with the prototype to complete specific tasks. This helps identify usability problems, gather user feedback, and assess whether the design meets user expectations and needs before final development.

Data collection tool: The study will be conducted using a Google Form survey. A useful method to gather quantitative data, in which I can receive genuine feedback on the implemented features and recommendations for future work.

Participants: Due to time constraints, we are only able to conduct the testing in this unit's workshop. In total, we were able to test with two groups (Group 96 and 47) and get responses from 5 participants.

Time Frame and place: This survey was conducted in-person at the unit's workshop on the 9th of May 2025.

User Tasks: During the usability testing session, participants were asked to complete a series of predefined tasks, which included:

- (1) Users are permitted to freely navigate throughout the application (Test visual elements and overall navigation)
- (2) Choose one of the predefined situations and interact with the AI (Test the situational roleplay feature and the AI)
- (3) Perform a translation and pronunciation during the interaction (Test translation and pronunciation feature)
- (4) End the interaction and evaluate the result, feedback, and recommendations (Test system feedback accuracy)
- (5) Create a custom roleplay scenario (Test the situational roleplay feature and the AI)
- (6) Interact with the newly created scenario (Test the custom situational roleplay feature and the AI)

Consent and confidentiality: All information collected during this usability test will be kept strictly confidential and used solely for the purpose of evaluating the prototype's usability. Testers are asked to provide informed consent to the analysis of their interactions with the prototype. No personal information will be shared or published. Tester feedback will be anonymized and handled under ethical research standards.

Data Analysis

1. Participants' demographics

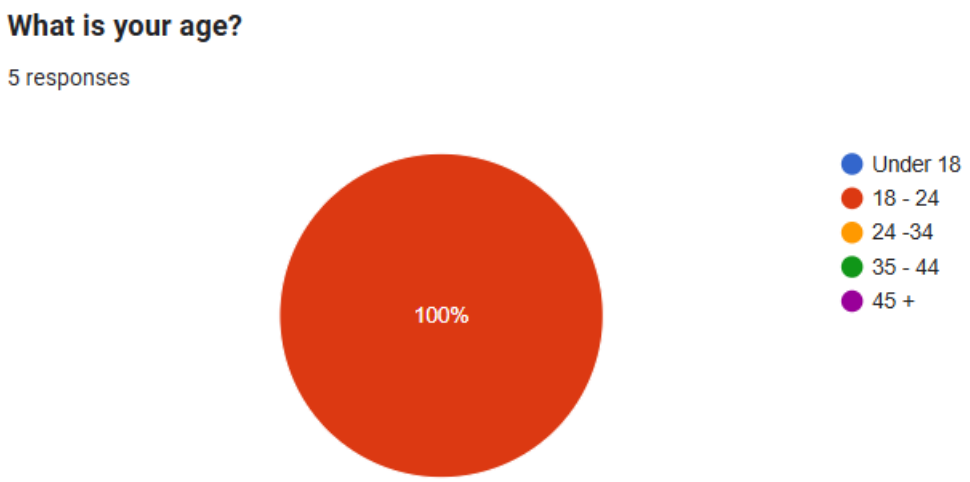


Figure 3: Pie chart showing participants' age range

- Age Distribution: 100% aged 18-24

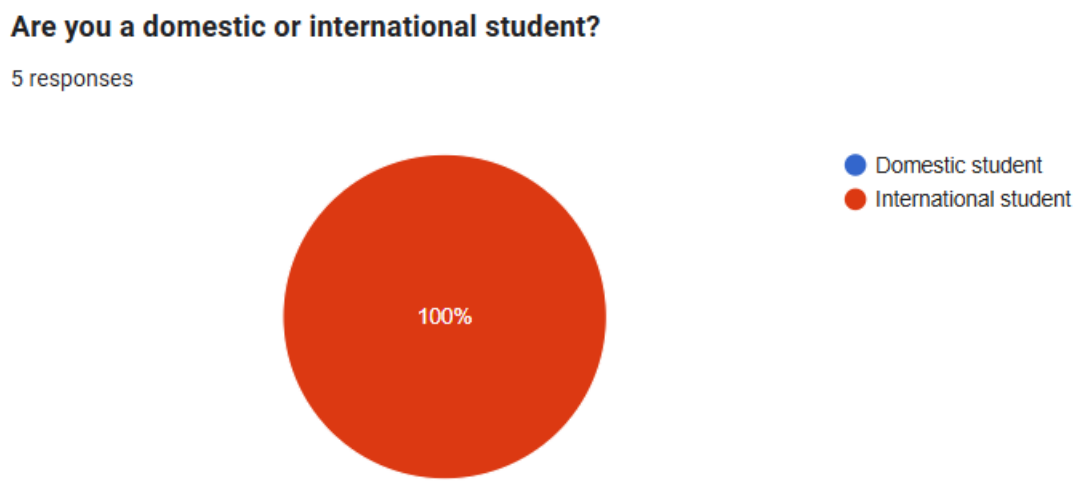


Figure 4: Pie chart showing participants' residency status

- All the testers are international students

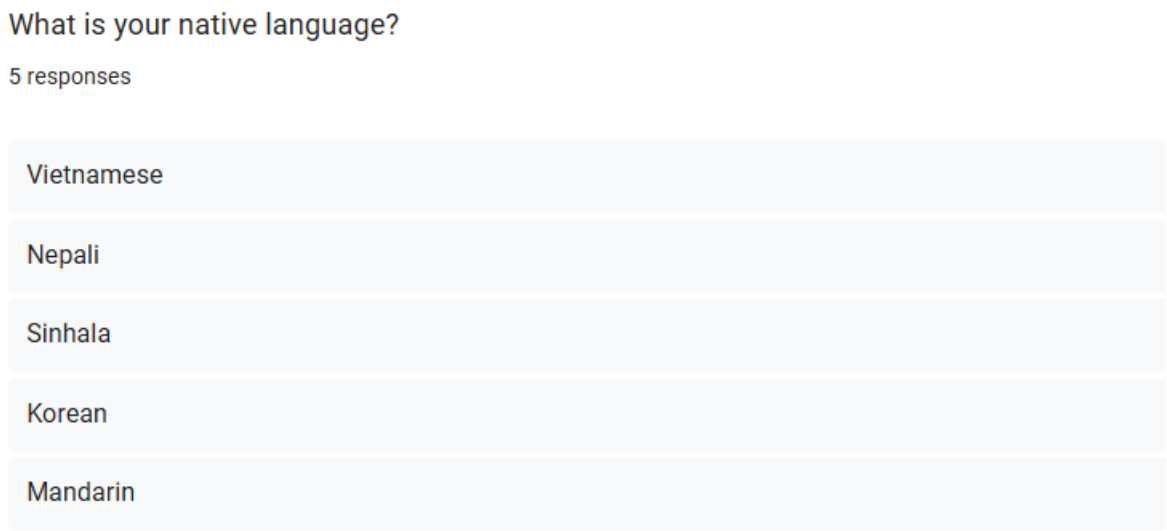


Figure 5: Particpants' native language

- Language diversity of testers:
  - Vietnamese
  - Nepali
  - Sinhala
  - Korean
  - Mandarin

How long have you been living in your current country of study?

5 responses

2
2 years
Few months
half a year
1 year

Figure 6: Participants' current duration of stay in Melbourne

- Duration of stay in Melbourne
  - 2 years
  - A few months
  - Half a year
  - 1 year

Conclusion: The testers are young adults aged 18-24. All are international students (target audience) from diverse backgrounds who are staying in their current country of study for varying lengths of time.

2. Application Interface

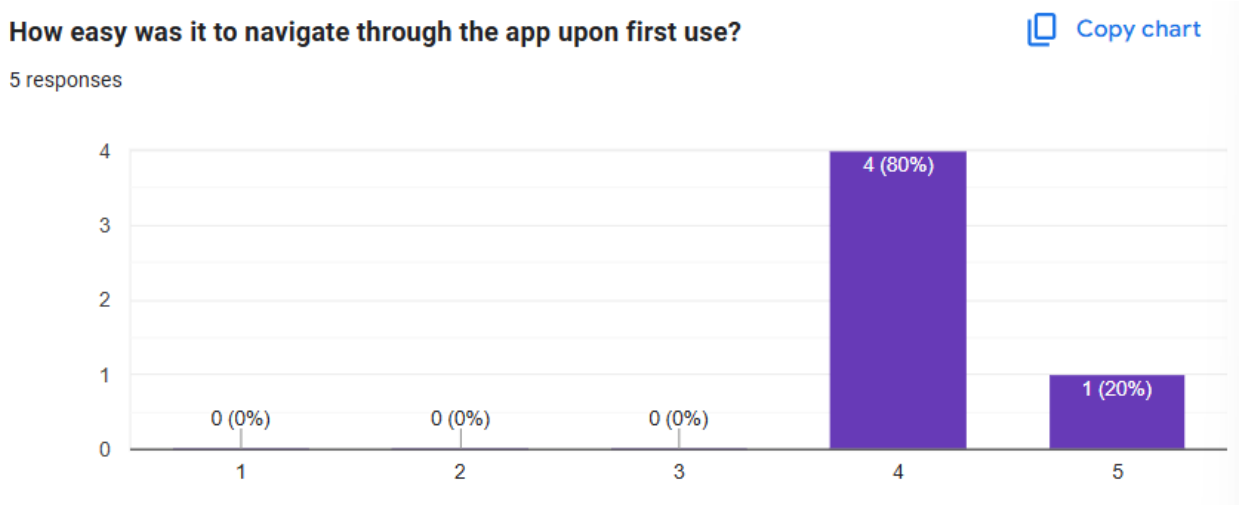


Figure 7: Bar chart showing the ease of navigation

- Ease of first-time use
  - Moderate Easy (80%)
  - Very Easy (20%)

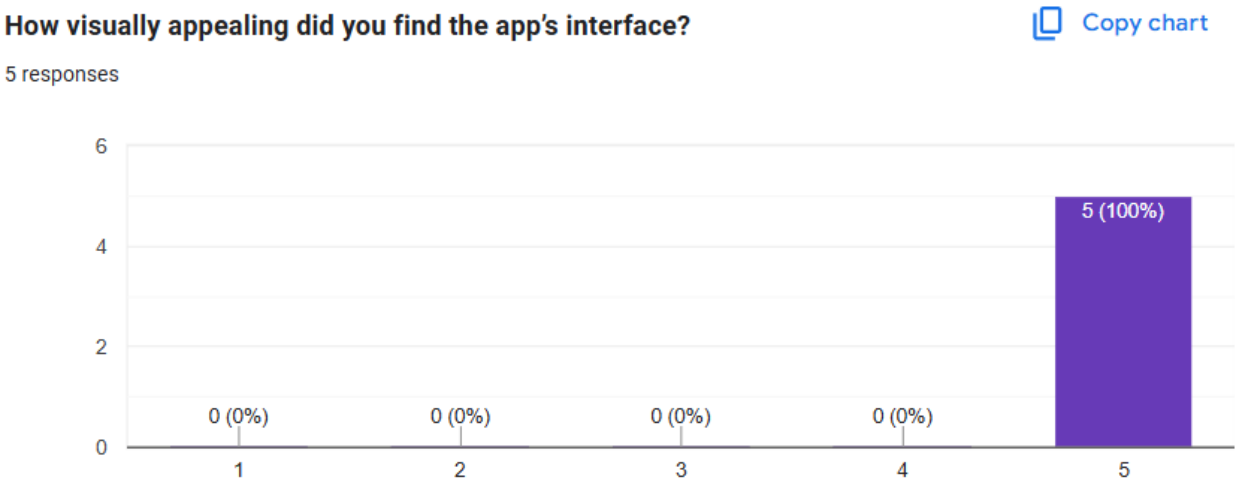


Figure 8: Bar chart showing participants' rating on interface

- A visually appealing interface
  - Very Appealing (100%)

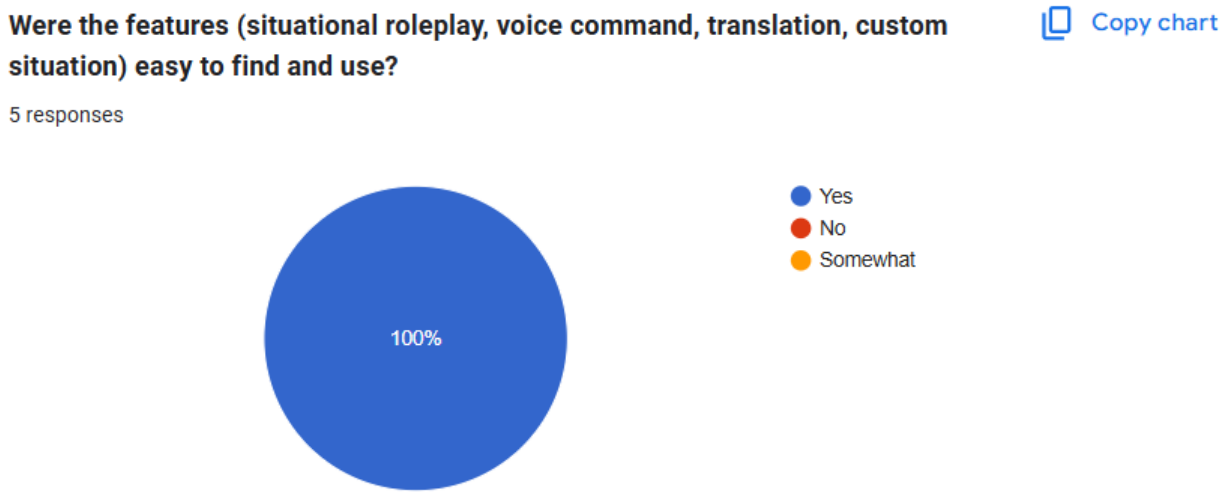


Figure 9: Pie chart showing rating on ease of finding the features

- Navigating functions
  - Yes (100%)

Conclusion: The application interface received positive feedback from testers, who found it visually appealing and easy to navigate. Features were reported to be easily accessible, contributing to a smooth and user-friendly experience.

3. Main Feature: Situational Roleplay

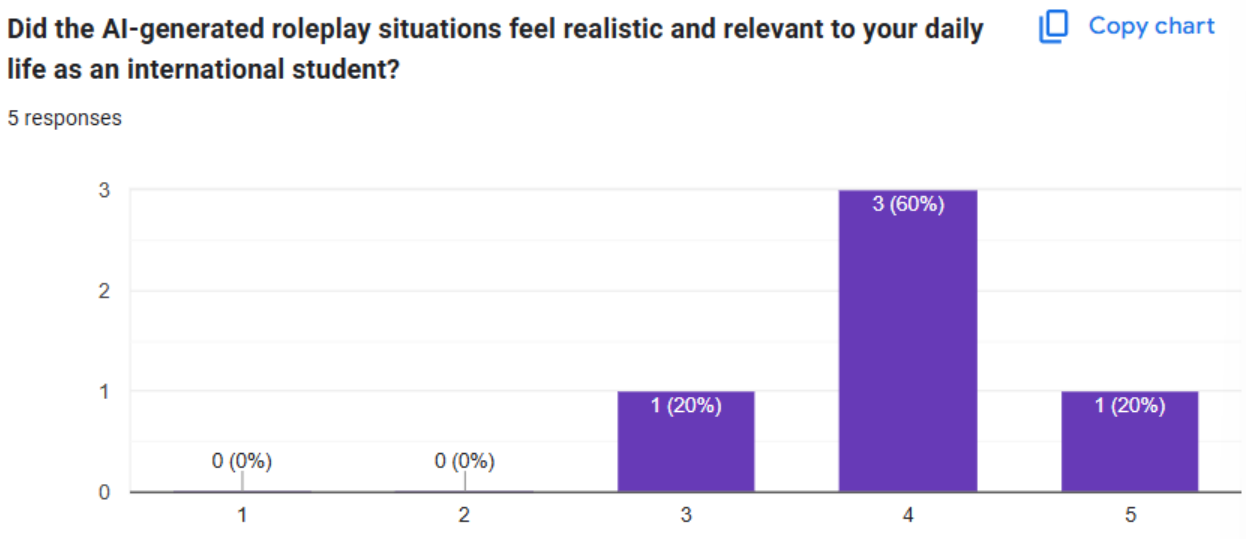


Figure 10: Bar chart showing rating of how realistic and relevant the scenarios are

- The AI scenario feature relevancy:

- Normal (20%)
- Quite useful (60%)
- Extremely useful (20%)

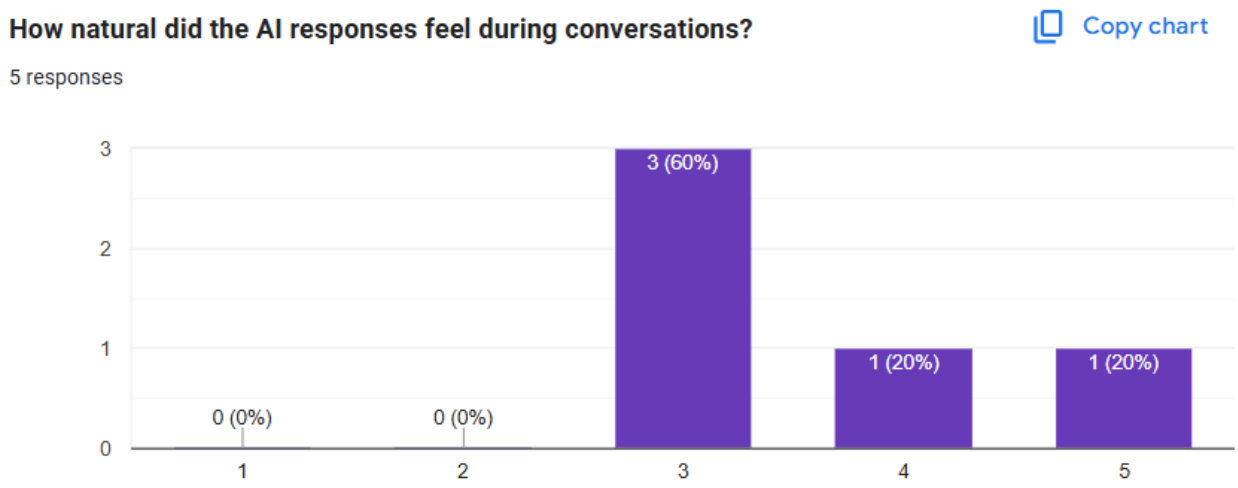


Figure 11: Bar chart showing participants' rating on the AI tone

- Realness of AI responses
  - Neutral (60%)
  - Mostly Natural (20%)
  - Very Natural (20%)

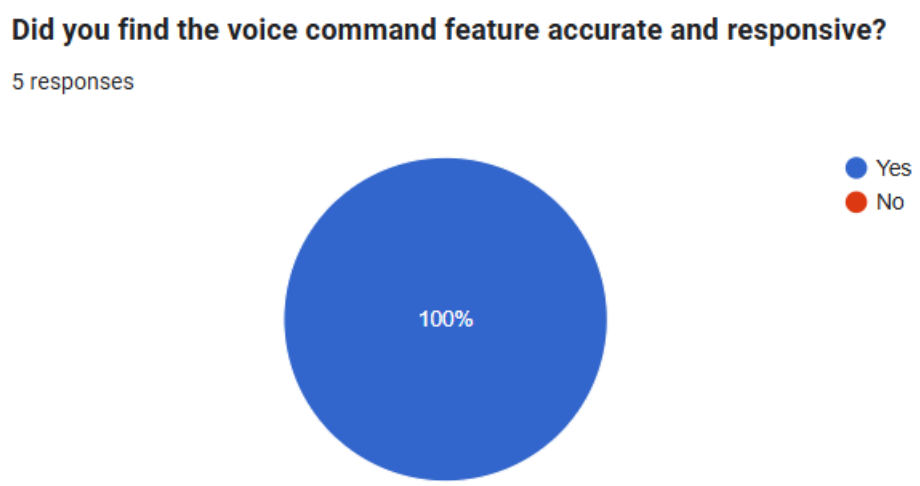


Figure 12: Pie chart showing rating of the speech-to-text display

- All testers found the voice command feature accurate and responsive

Describe any issues you faced with the voice command feature (if any)

1 response

Just minor inconsistence with the text recorded

Figure 13: Addtional errors

- One tester does note a minor inconsistency with the voice command

Conclusion: The Situational Roleplay feels realistic and similar to daily interactions. Most testers found the AI responses to strike a balanced tone—neither overly artificial nor entirely natural—while a few noted a more human-like quality. The speech-to-text display was generally accurate for most testers, with only one tester reporting minor discrepancies, none of which resulted in significant errors.

4. Feedback and Learning Screen

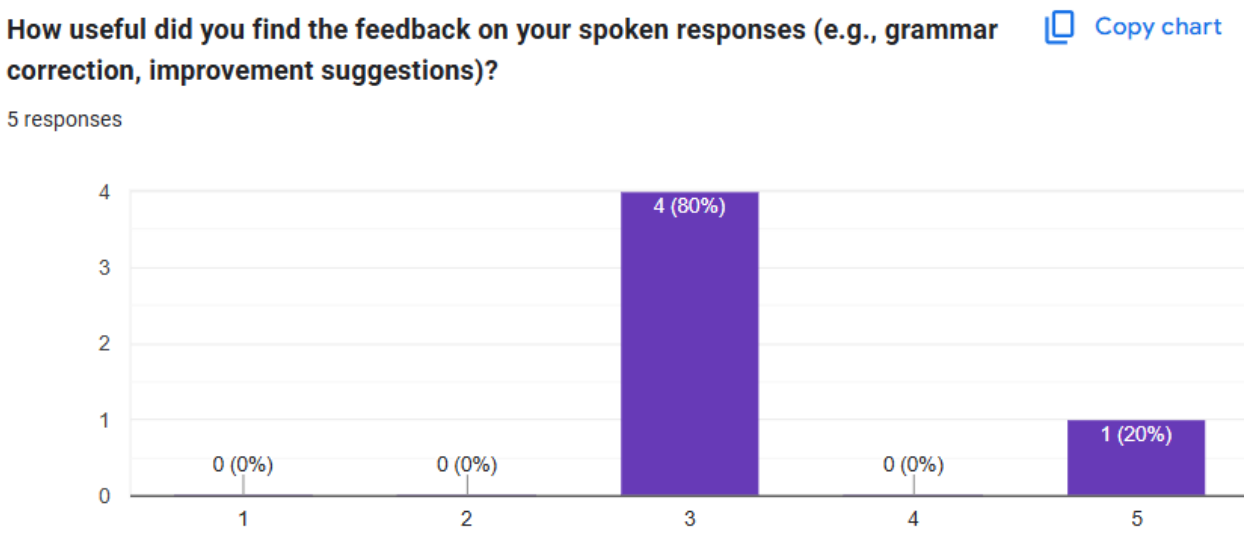


Figure 14: Bar chart of participants' rating the feedback on each responses

- Feedback Helpfulness:
  - Neutral (80%)
  - Extremely Useful (20%)

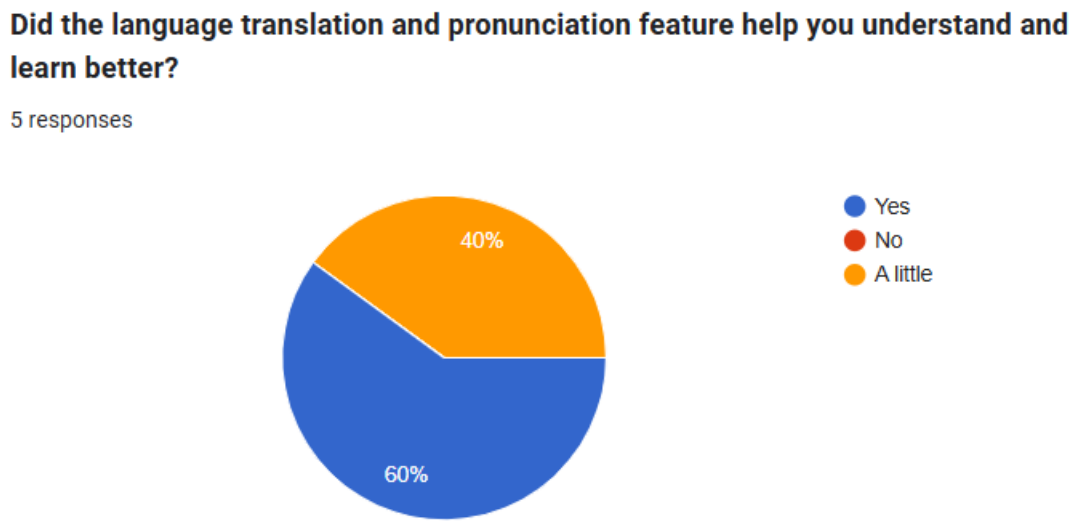


Figure 15: Pie chart showing rating on the usefulness of the translation and pronunciation feature

- Translation and Pronunciation helpfulness
  - Yes (60%)
  - A little (40%)

Conclusion: Most testers found the feedback and recommendation feature to be neutral, while one tester considered it highly beneficial. Overall, the translation and pronunciation features were found to be helpful during testing.

5. Custom Scenario

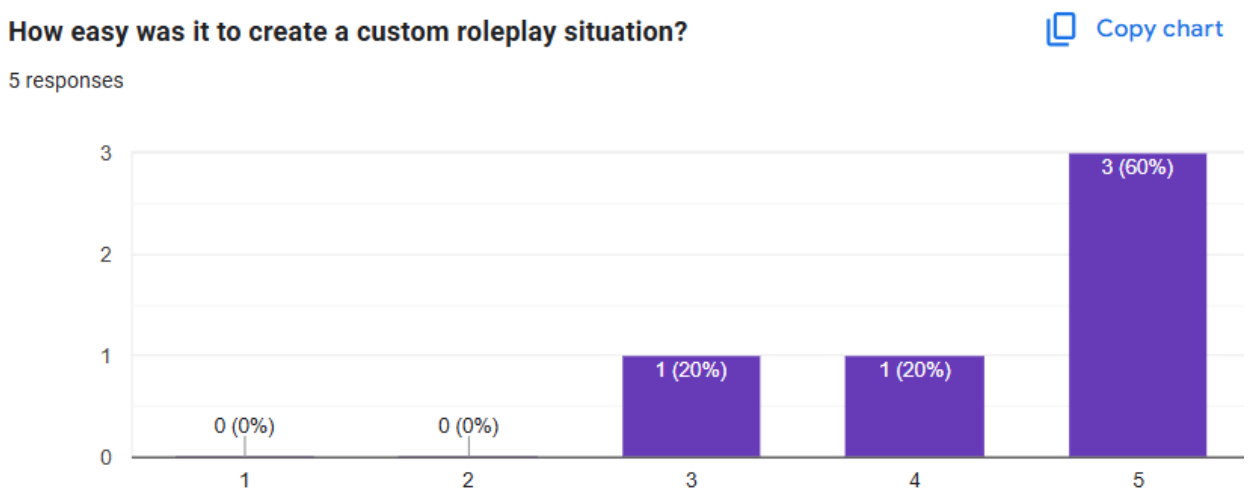


Figure 16: Bar chart rating how easy to navigate the custom scenario feature

- Ease of creating a custom scenario:
    - Neutral (20%)
    - Moderately Easy (20%)
    - Very Easy (60%)

Conclusion: Testers found customizing their unique scenario was straightforward

6. Overall Satisfaction

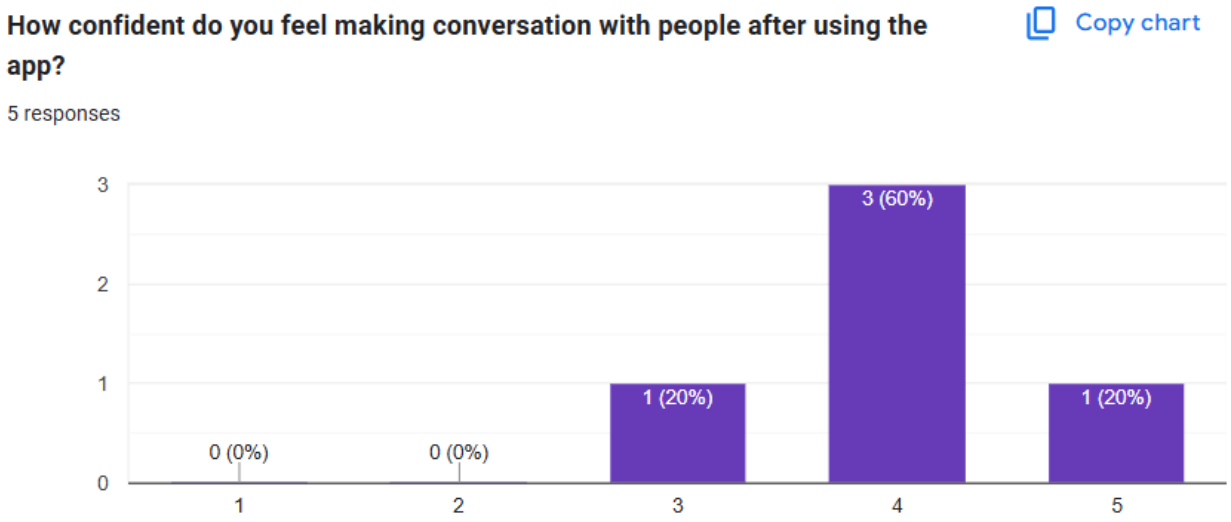


Figure 17: Bar chart showing participants' rating on confidence post-usage

- Confident boost after usage:
    - Neutral: 20%
    - More confident: 60%
    - Extremely confident: 20%

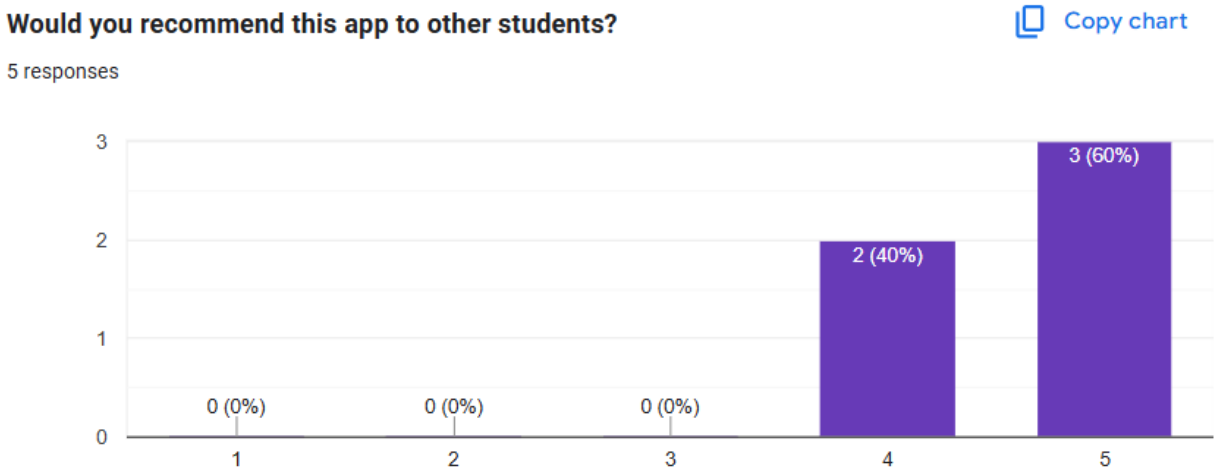


Figure 18: Bar chart showing recommendation probability

- Recommend to other students:
  - Considered (40%)
  - Definitely (60%)

Conclusion: The app displays a positive impact on testers' confidence, with the majority feeling more confident in starting a conversation. Only one participant felt neutral, and none reported decreased confidence. In terms of recommendation, all testers responded positively, with the majority likely recommending the app to other students. These results suggest the app is effective in enhancing user confidence and is well-regarded among its testers.

7. Suggestions and Improvements

What was your favorite feature of the app, and why?

5 responses

My favorite feature is the role feature as it allow me to be in the correct situation

Conversation and feedback

Custom bot

Grammar correction. Because I can notice bad habits.

The realistic AI generated situational roleplay. I can test my conversation skills and get better in the comfort of my own home.

Figure 19: Participants' favorite feature

What feature(s) did you find least useful or most confusing?

5 responses

I that the roleplay feature at first was confusing but overall using it make it easier to use

Not sure

Everything was working well

Creating a role play

For me the least useful is the translation and pronunciation feature, cause I have not problem with translation or proper pronunciation

Figure 20: The least useful feature

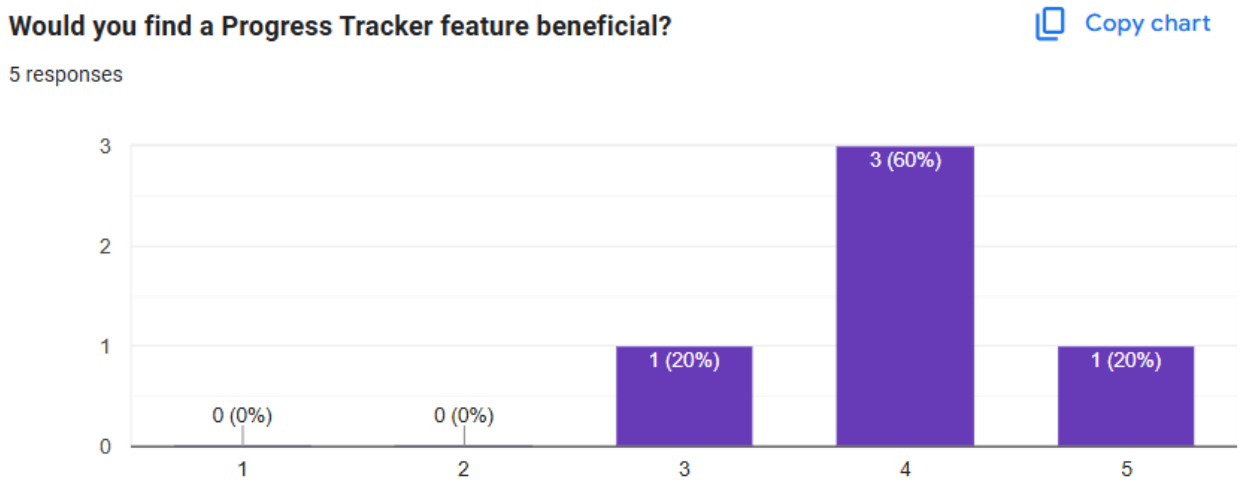


Figure 21: Bar chart display percentage of support for the upcoming Progress Tracker feature

- The benefits of a Progress Tracker feature
  - Neutral (20%)
  - Somewhat beneficial (60%)
  - Very beneficial (20%)

What improvements or new features would you suggest for the app?

3 responses

A tutorial introduction to the app feature would be great

I would suggest adding more system languages so that other my friends can use it as well.

More languages would be nice, some of my friends are visiting taiwan and it would be nice to for them to practice a casual conversation in mandarin.

Figure 22: Additional suggestion from participants'

Conclusion: Most testers value the scenario roleplay feature of the app, with several responses highlighting the realistic practice opportunities and improving conversation skills. One of the participants also mentioned that the grammar correction feedback (a minor feature) was helping them correct bad habits.

The Progress Tracker received mostly positive feedback, with 80% of respondents finding it beneficial, 20% were neutral, and none rated it negatively. One tester initially found the roleplay feature confusing, though they eventually recognized its value. The translation and pronunciation

feature (minor feature) was considered the least useful by one user who already felt confident in that area.

For future development, testers suggested three main improvements: adding a tutorial introduction to help new users navigate the roleplay scenarios, expanding language support for roleplay scenarios, and implementing Progress Tracker functionality.

VI. Key Findings and Future Implementations

From the data pulled from the user testing, I have concluded some valuable insights that significantly influence the direction for future iterations of this prototype. Overall, the MojiMoji was well-received; testers consistently praised the intuitive interface, realistic AI roleplay scenarios, and the possible learning opportunities.

1. User Experience and Interface:

Findings:

- The interface was described as visually appealing and intuitive. Navigation was easy, with testers able to access features without confusion. Customizing unique scenarios was straightforward.
- One tester reported experiencing minor confusion on how the AI roleplay feature works

Future Implementations:

- Maintain the current UI structure but enhance first-time user experience by introducing an onboard tutorial to explain the roleplay flow and feature use.

2. Core Feature: Roleplay Scenario

Findings:

- The situational roleplay was well-received; testers highlighted its realism, mimicking daily conversations. The AI responses were neutral in tone – neither too robotic nor too human. The speech-to-text was generally reliable, with one minor complaint about its accuracy.
- The Customize Scenario was praised as it gave users a more personalized experience with a suitable difficulty level that matches their levels and further enhances the overall realism.
- One user stated, “More languages would be nice, some of my friends are visiting Taiwan, and it would be nice for them to practice a casual conversation in Mandarin”. Our application system supports many languages; however, the

AI can only make conversation in English, meaning it would only be beneficial for an English-speaking environment.

Future Implementations:

- Continue to refine the AI dialogue system to have more natural language flow, and polish the speech recognition for consistent performance.
- Enhance the AI capability to support multiple language interactions. Making the app more effective for a non-English-speaking environment. The first major move to push this app onto the international market.

3. Supportive Tools and Features

Findings:

- Translation and Pronunciation tools were helpful to most testers, although one tester found them the least useful
- The overall Feedback and Recommendation feature fell short of our expectations, with most testers feeling it was okay and somewhat beneficial, with one user reporting that the grammar correction helped with bad habits.
- The introduction of a possible Progress Tracker feature was well received, with 80% of the testers reporting support for the feature and its future benefits

Future Implementations:

- Continue to improve the clarity of the feedback and fix the suggestions to be more personalized and actionable.
- Implement the Progress Tracker to improve clarity, increase engagement, and motivation.
- Implement a tutorial to explain the Translation and Pronunciation feature so that users can get the most use out of it

Conclusion: The key findings above reflect that the prototype has a strong foundation with clear areas for improvement. Based on the feedback from our testers, our team has decided to prioritize these iterations:

- A tutorial for new users
- Broader language support for roleplay scenarios
- Implement Progress Tracker

VII. Future Implementation: First Look at the Design

Expanded AI Language Support

- **Feedback**: Usability testing revealed participants with diverse language backgrounds (Vietnamese, Nepali, Sinhala, Korean, Mandarin). One user mentioned a use case of the application outside a non-English-speaking country.
- **Current solution**: Expanded from English-only in the initial version to now supporting Chinese and Japanese
- **Next implementation**: Additional language support planned for Vietnamese, Hindi, German, Spanish, and more
- **Expected Impact**: Broadened user base by providing native language learning support to a wider range of international students

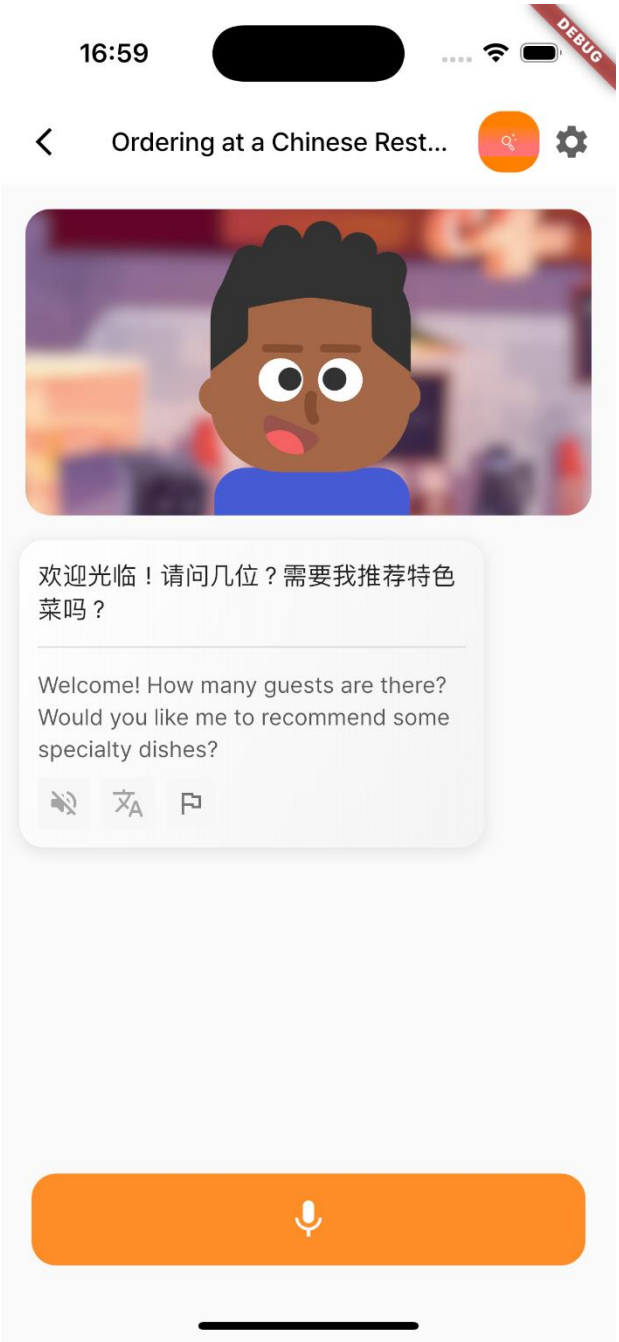
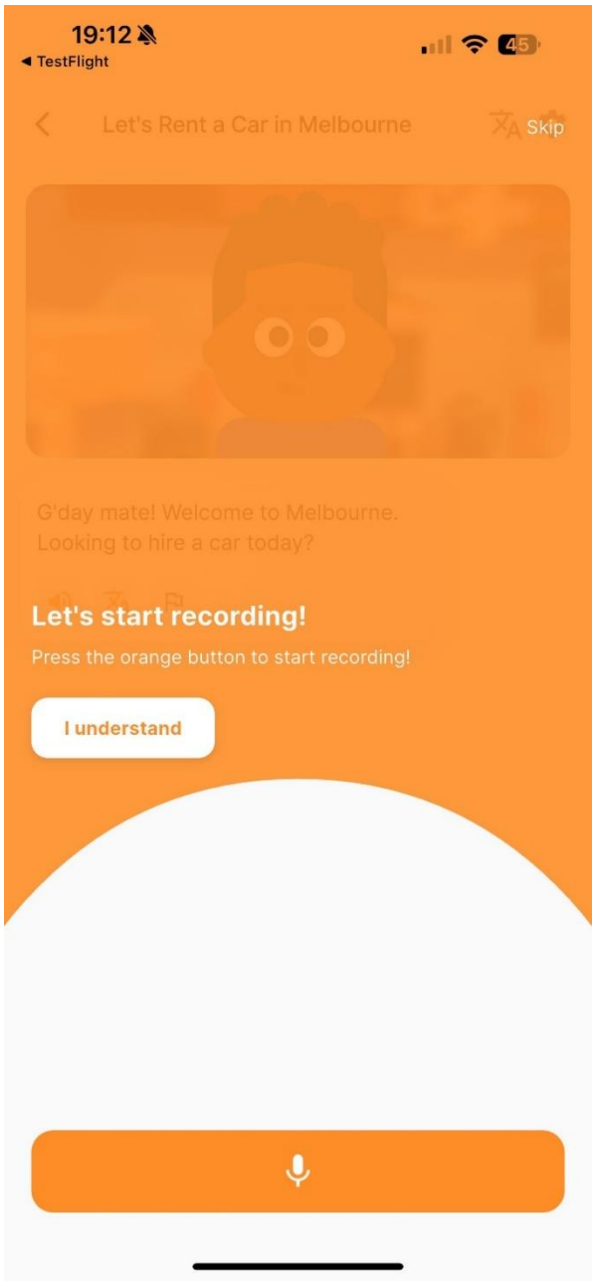


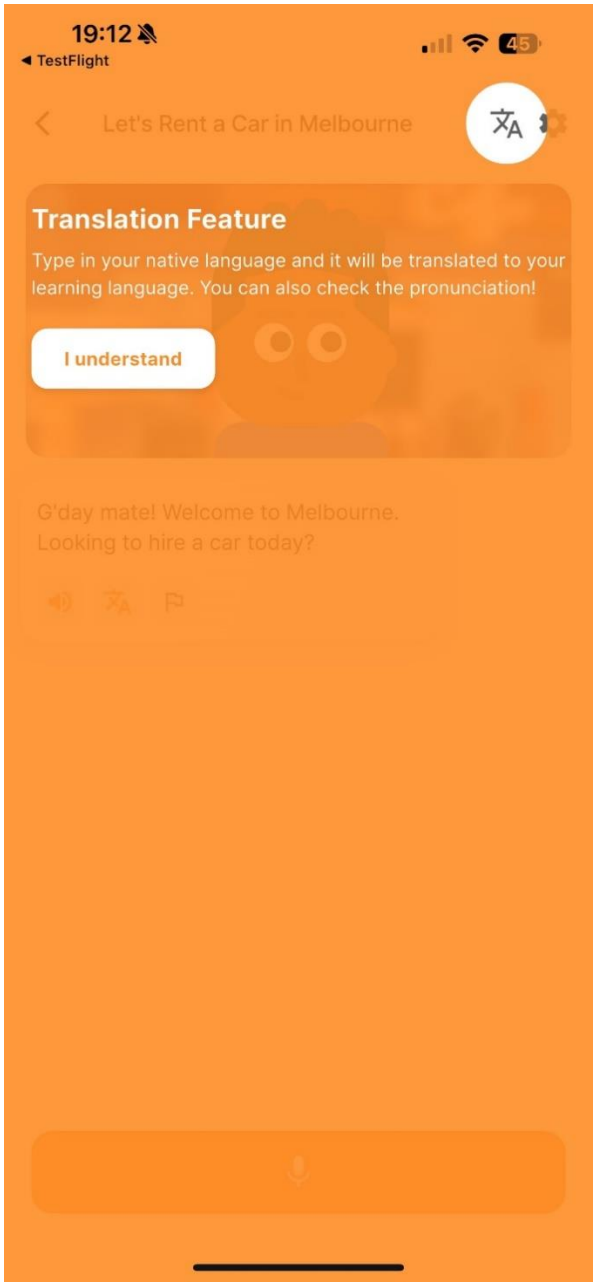
Figure 23: AI making conversation in Chinese

Implemented Tutorial System

- **Feedback**: One test participant had initially struggled to understand how to use the recording features
- **Current Solution**: Step-by-step interactive tutorial system for new users

- **Next implementation**: Adding contextual hints and developing additional tutorials for advanced features
- **Expected Impact:** Improved first-user experience, lowered barrier to entry, and reduced app abandonment


19:12
TestFlight
Let's Rent a Car in Melbourne
Skip
G'day mate! Welcome to Melbourne.
Looking to hire a car today?
Let's start recording!
Press the orange button to start recording!
I understand


19:12
TestFlight
Let's Rent a Car in Melbourne
Translation Feature
Type in your native language and it will be translated to your learning language. You can also check the pronunciation!
I understand
G'day mate! Welcome to Melbourne.
Looking to hire a car today?

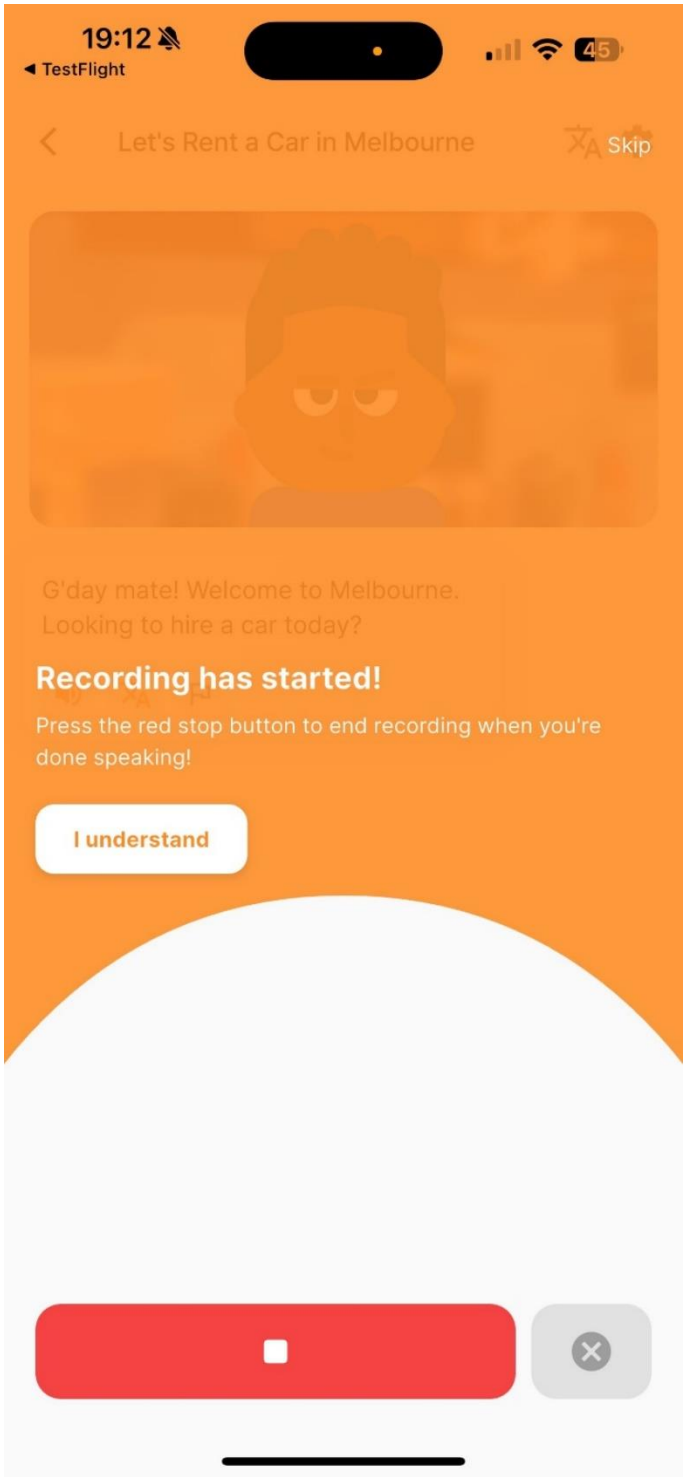


Figure 24: The tutorial introducing the scenario feature

Progress Tracking System

- **Feature description:** Comprehensive dashboard that visualizes and measures the user's language learning journey
- **Feedback**: 80% of testers indicated a progress tracker would be beneficial
- **Implementation**: Dashboard visualizing day streaks, daily speech activity, global score, and monthly activity.
- **Future Expansion Plans**:
  - Adding weekly activity analysis
  - Providing personalized learning recommendations
  - Achievement badge system based on progress
- **Expected Impact**: Enhanced user motivation and long-term learning engagement

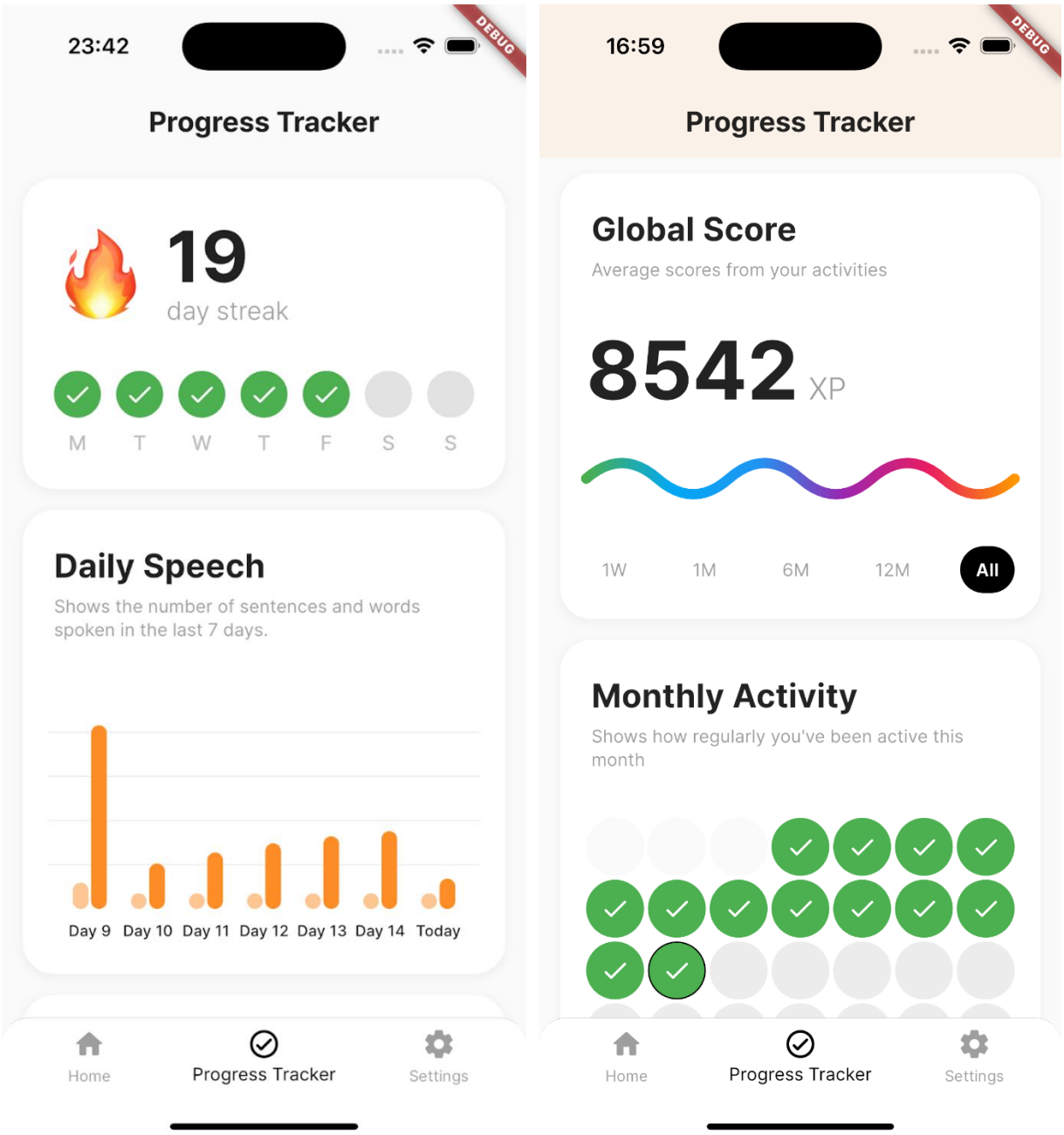


Figure 25: The Progress Tracker feature

VIII. Conclusion

MojiMoji was developed through an iterative, user-centered design process focused on realistic language learning via AI. The prototype evolved across three main iterations – enhancing scenario selection, adding custom scenarios, and improving AI interface elements. The design iterations were backed up with users' and team members' feedback.

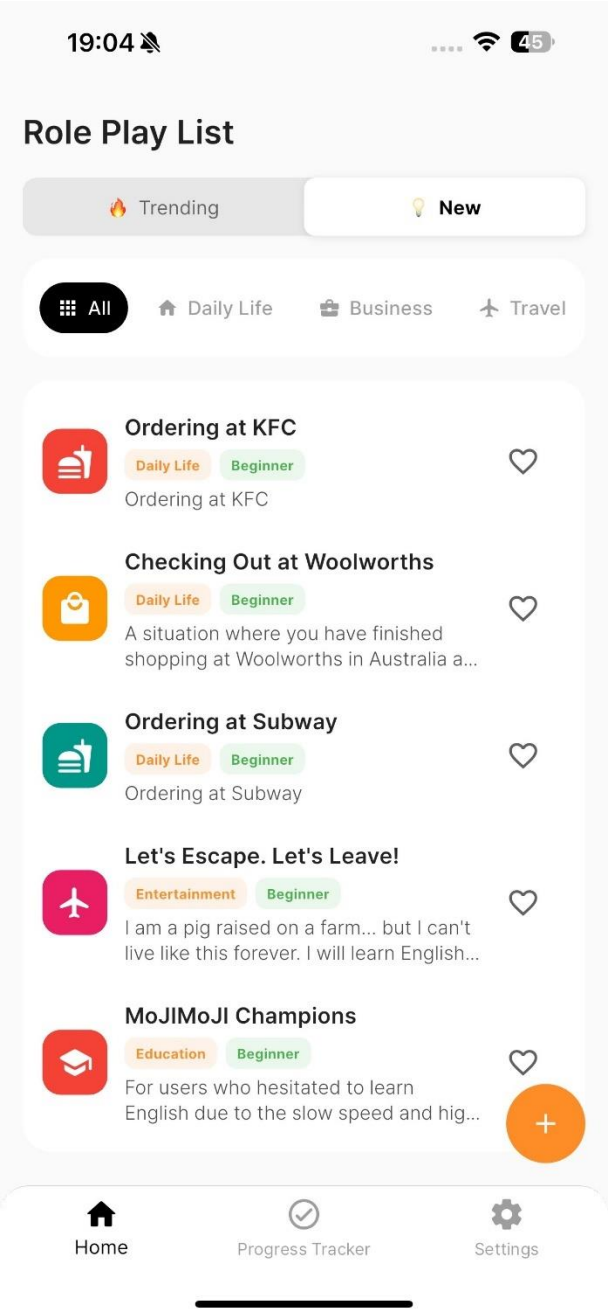
Usability testing with international students showed positive results, users found the app intuitive, engaging, and effective for building confidence in conversations. Scenario roleplay and customization were especially well-received, while some minor improvements were suggested for feedback clarity, speech-to-text accuracy, and language support. Future work will continue expanding on language support, refining AI responses and the Progress Tracker, and introducing further tutorials to guide new users.

IX. References

Figma. (n.d.). *What is Human Computer Interaction?* [online] Available at: 14 May 2025, https://www.figma.com/resource-library/human-computer-interaction/.

X. Appendix A: High Fidelity Prototype

Home Screen:



Roleplay Scenario Feature:

Choose a roleplay → Choose the AI voice → Start conversation

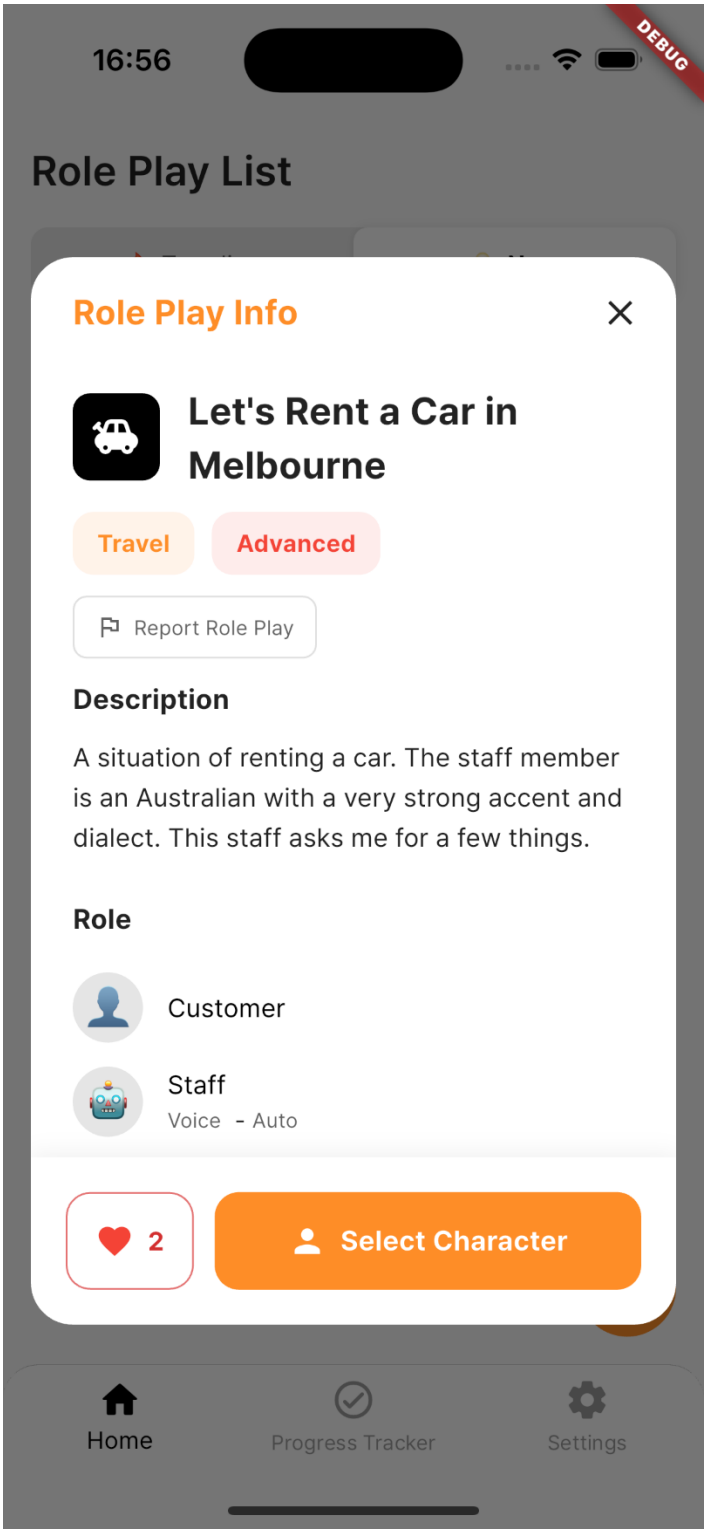


Figure 26: Choose a roleplay screen

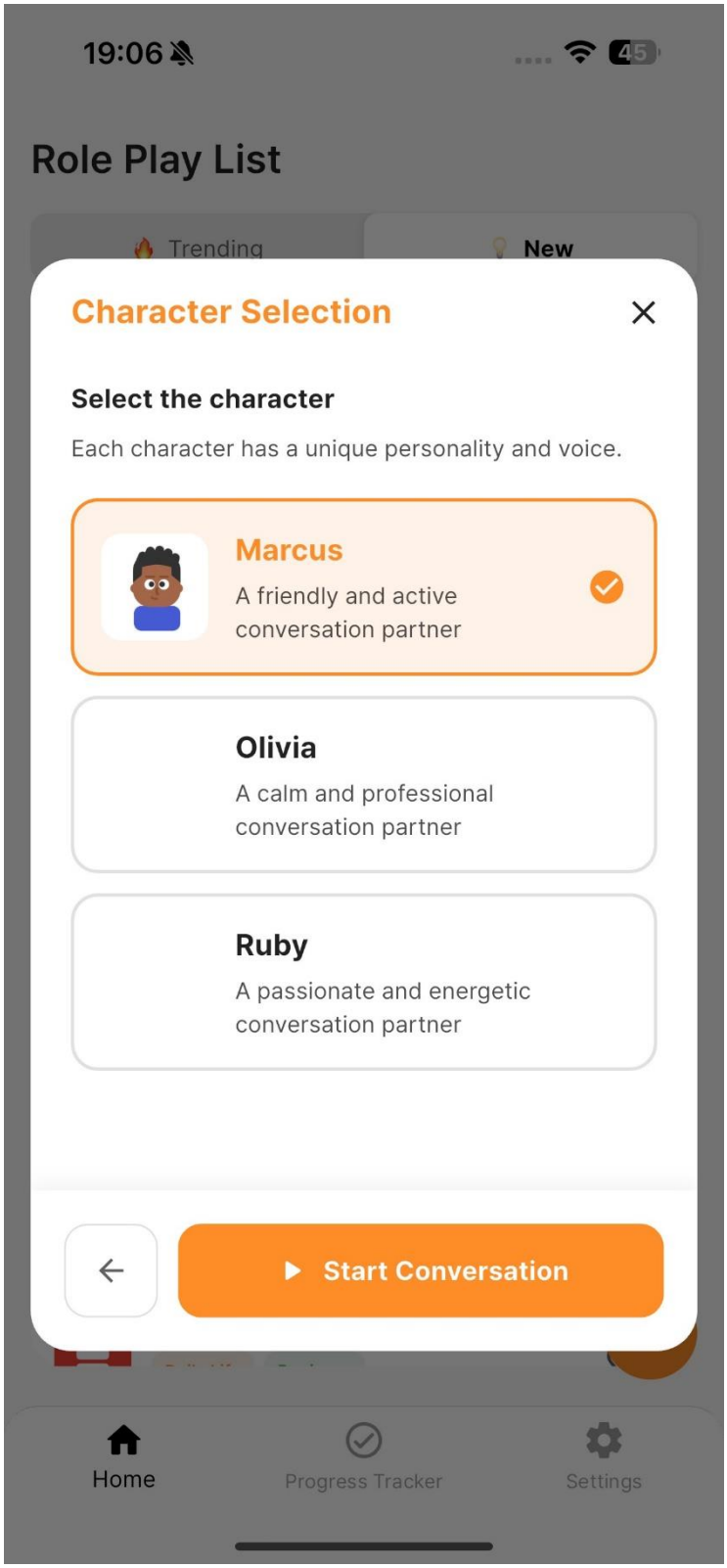


Figure 27: Choose the AI voice screen

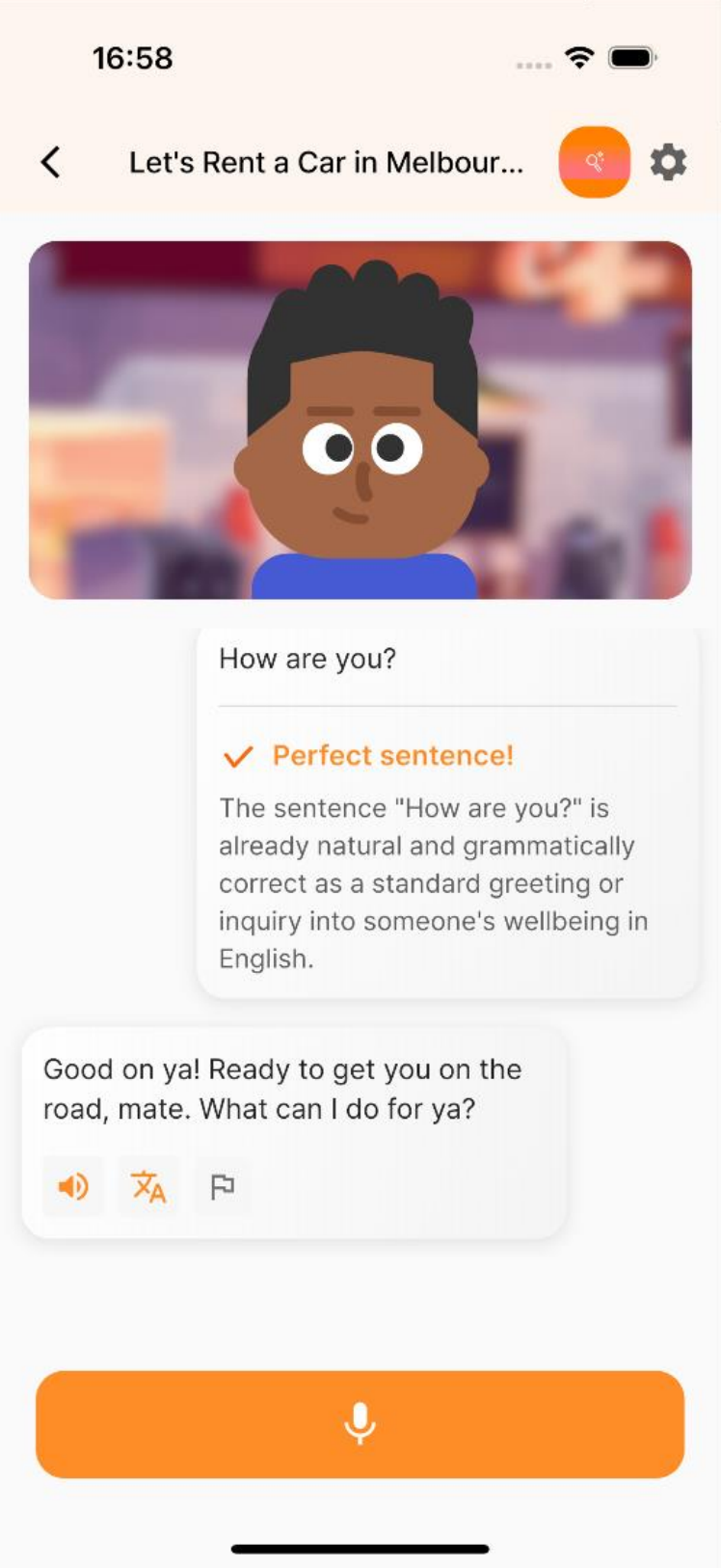
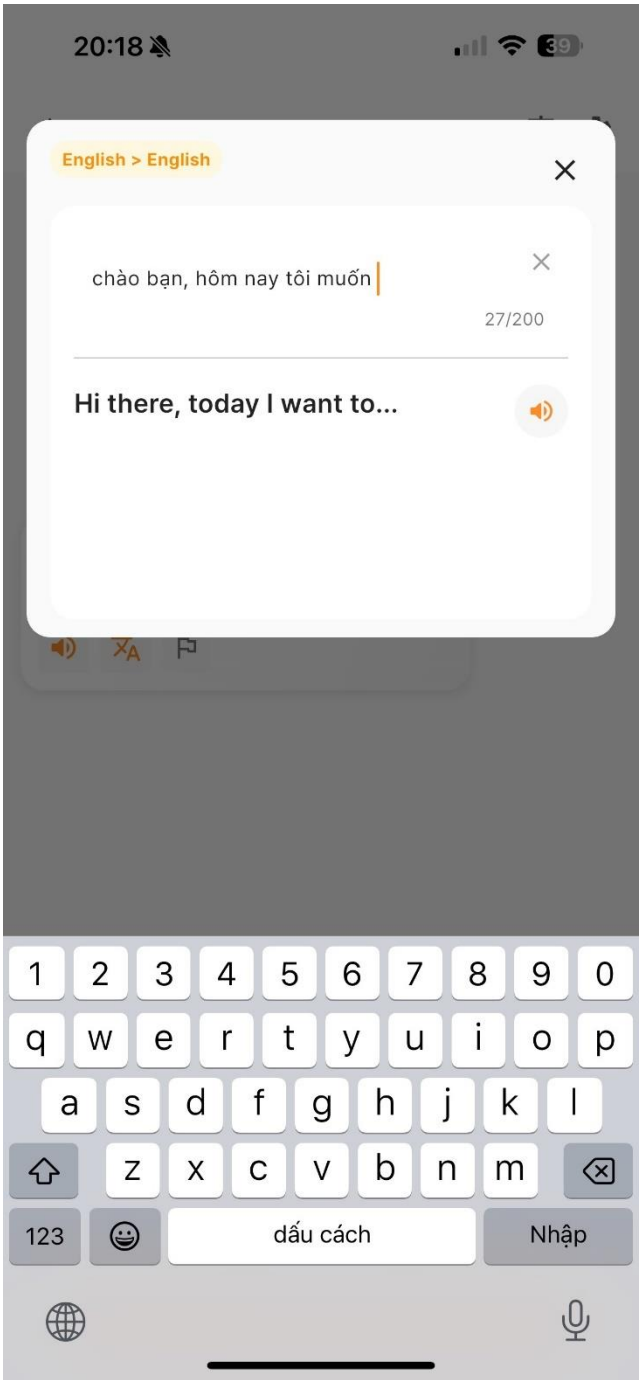
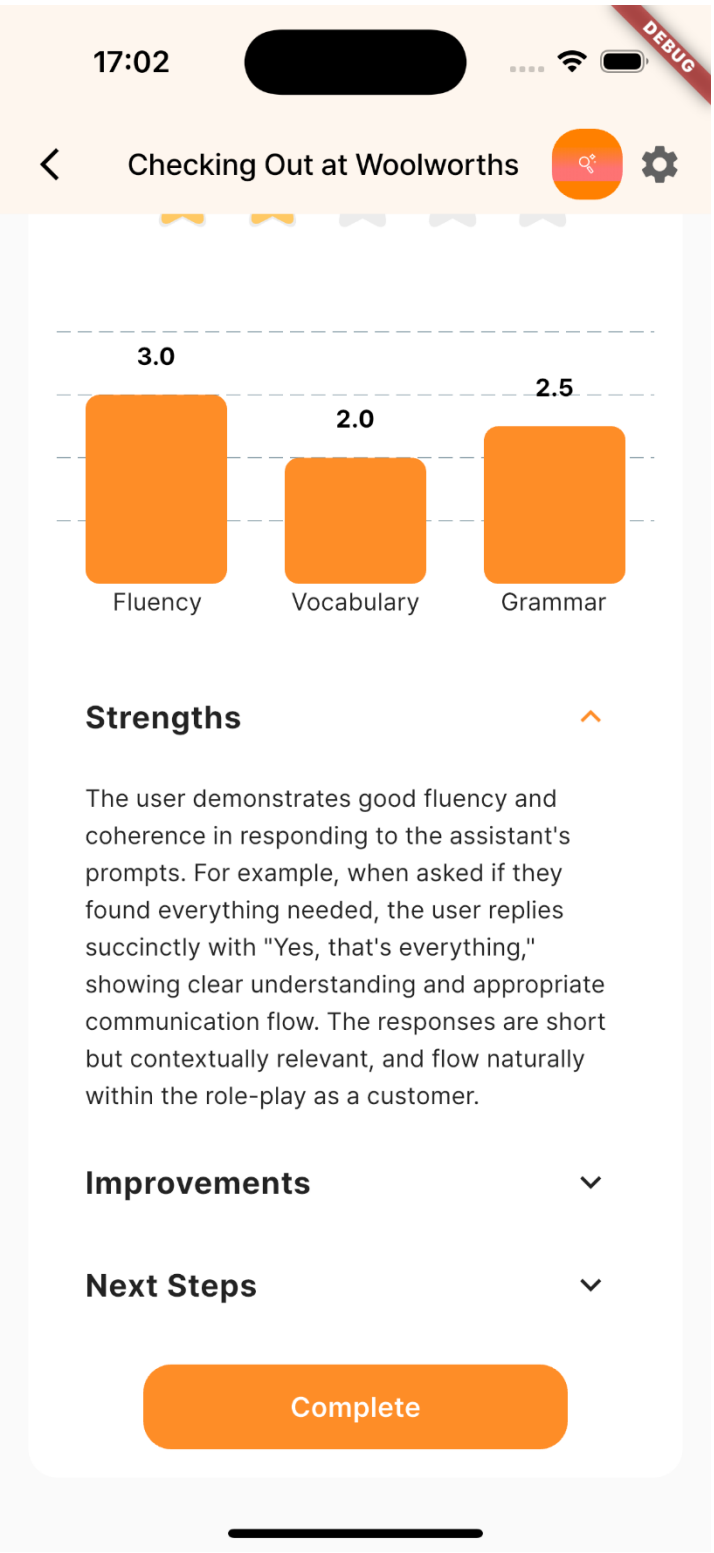


Figure 28: Start conversation screen

The translation and pronunciation feature:

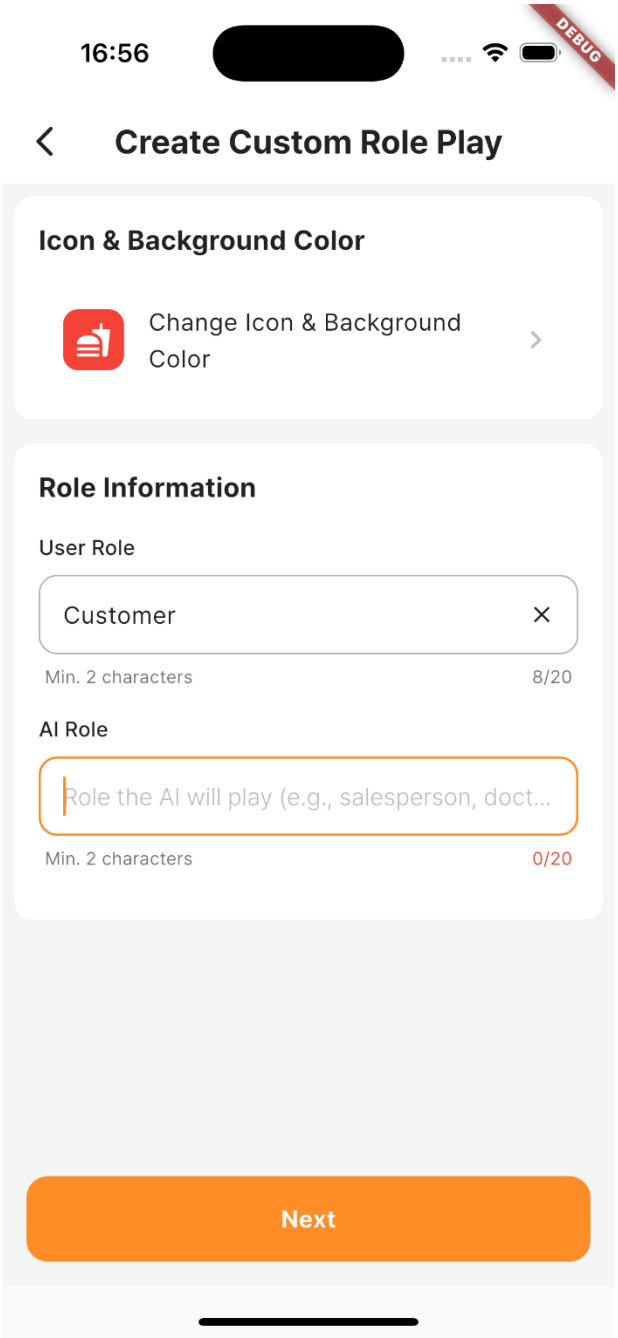


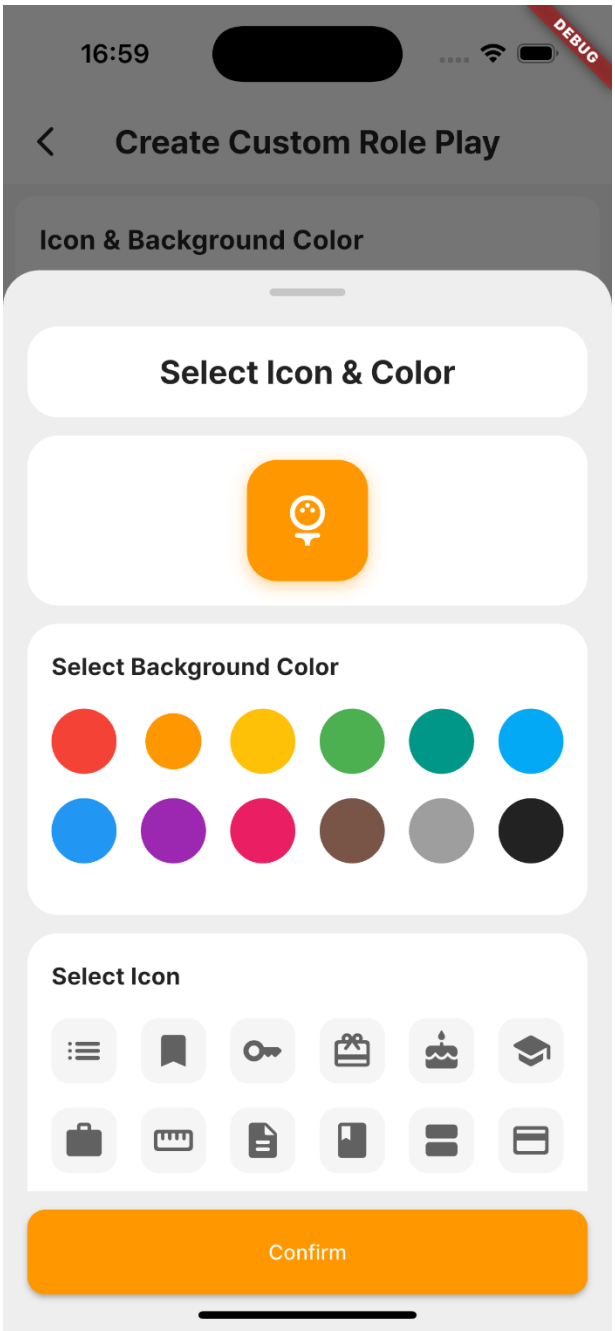
The feedback and recommendation screen:

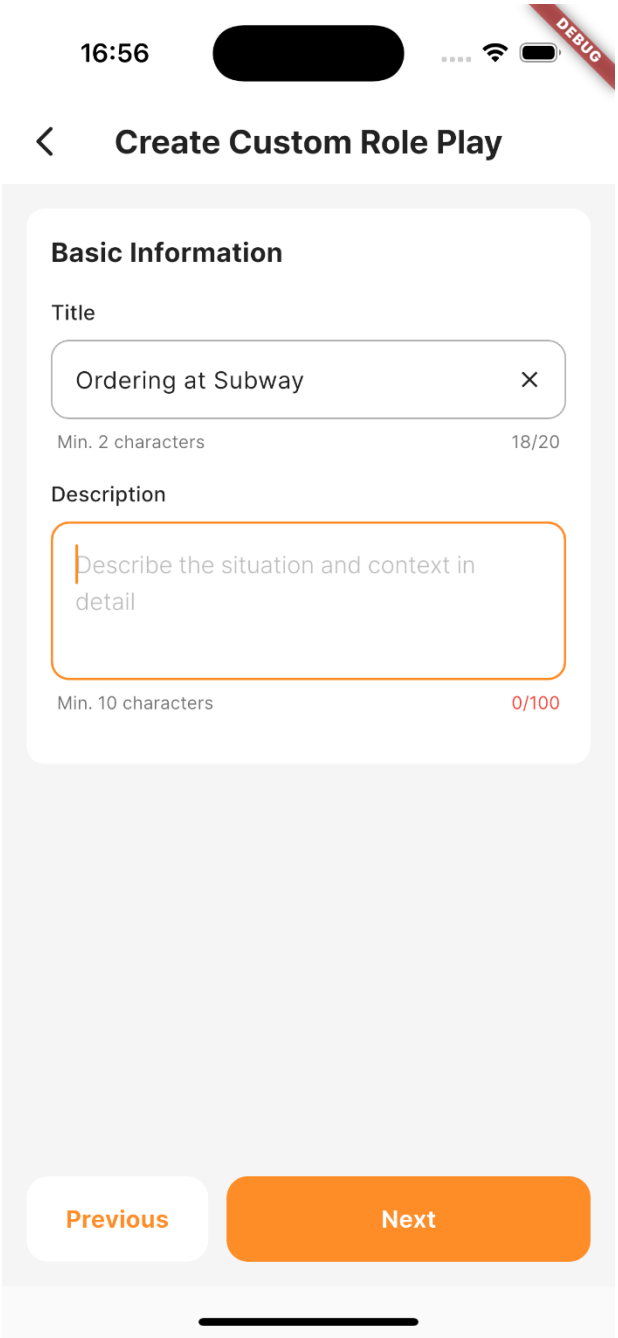

17:02
DEBUG
Checking Out at Woolworths
3.0
2.0
2.5
Fluency
Vocabulary
Grammar
Strengths
The user demonstrates good fluency and coherence in responding to the assistant's prompts. For example, when asked if they found everything needed, the user replies succinctly with "Yes, that's everything," showing clear understanding and appropriate communication flow. The responses are short but contextually relevant, and flow naturally within the role-play as a customer.
Improvements
Next Steps
Complete

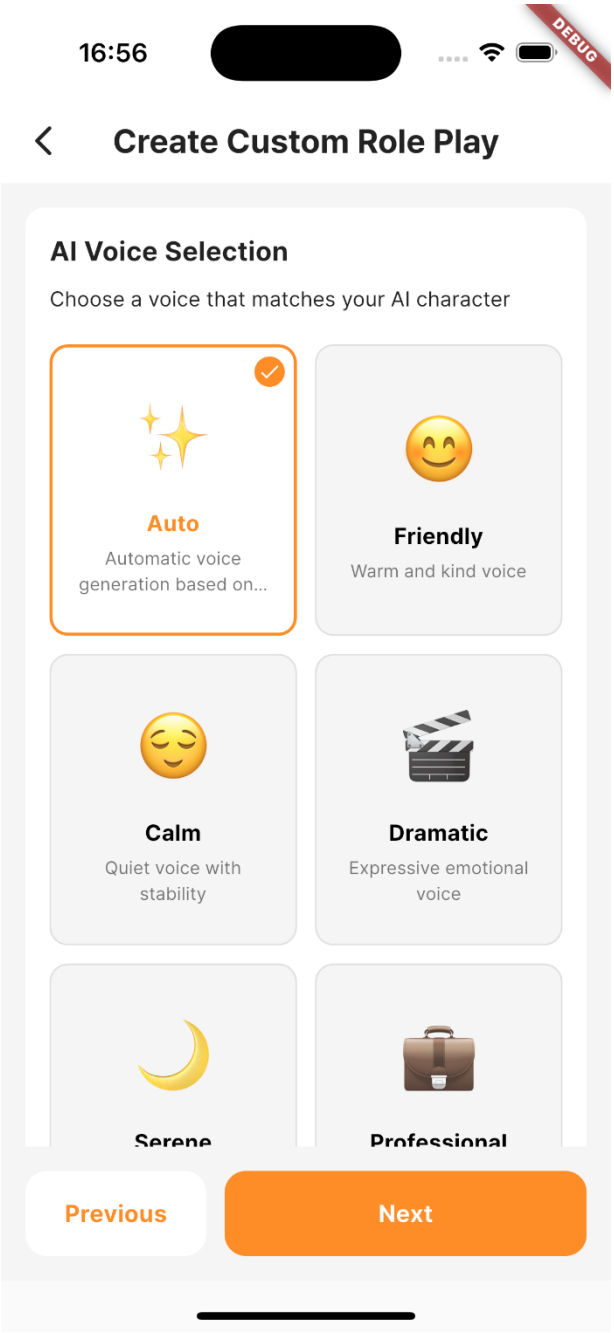
Customizing a Roleplay scenario:

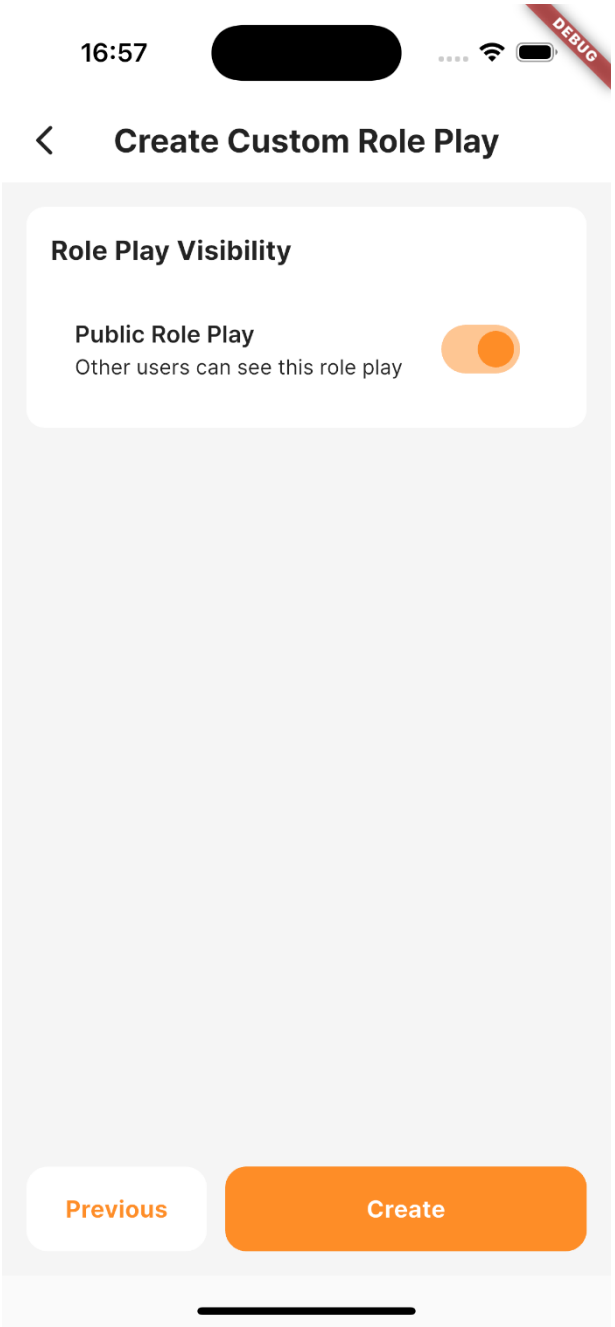
Choose the Icon and give a role to the user and AI → Give basic information about the scenario → Choose the AI voice → Decide the roleplay visibility → Scenario created successfully

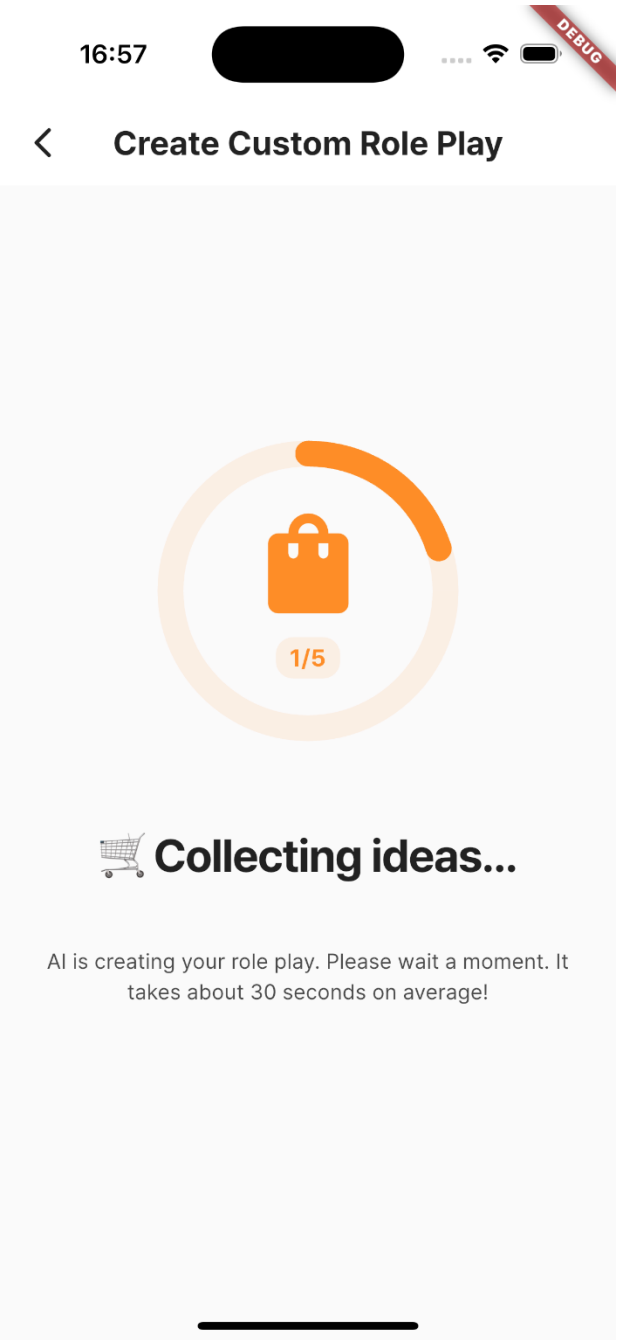

16:56
DEBUG
Create Custom Role Play
Icon & Background Color
Change Icon & Background Color
Role Information
User Role
Customer
Min. 2 characters
8/20
AI Role
Role the AI will play (e.g., salesperson, doct...
Min. 2 characters
0/20
Next

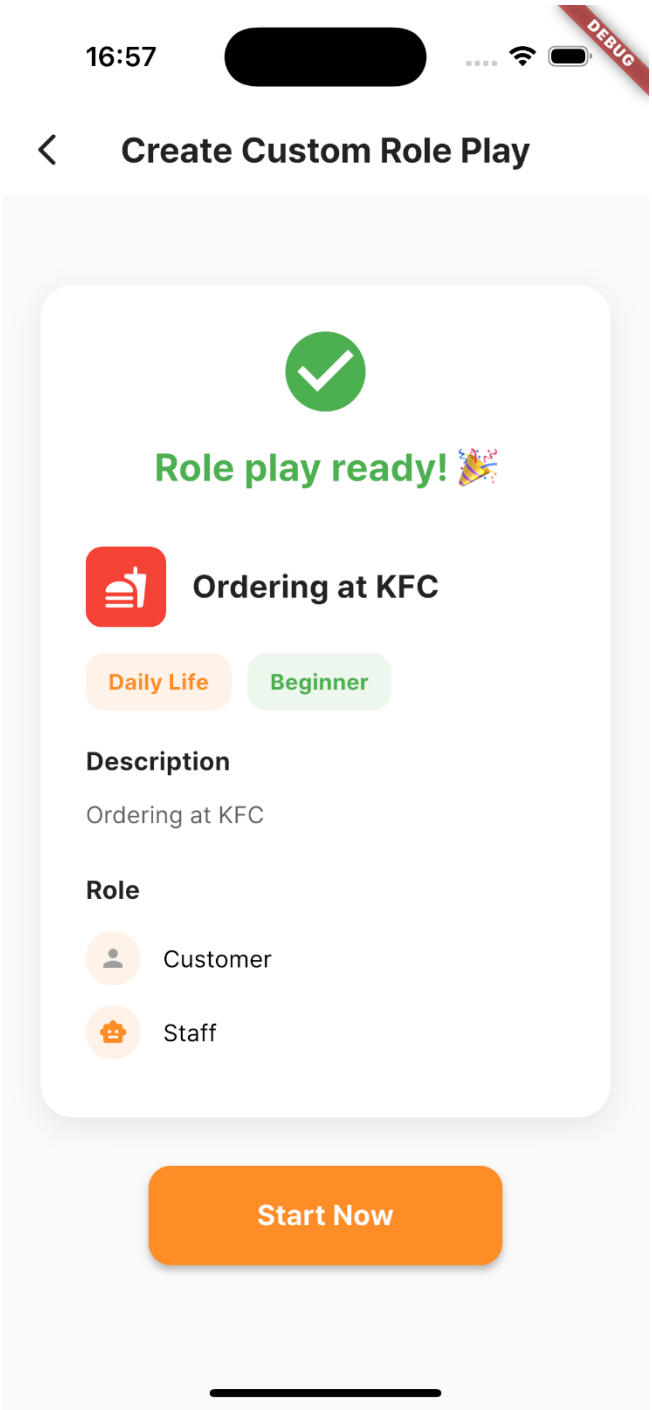

16:59
DEBUG
Create Custom Role Play
Icon & Background Color
Select Icon & Color
Select Background Color
Select Icon
Confirm


16:56
DEBUG
Create Custom Role Play
Basic Information
Title
Ordering at Subway
Min. 2 characters
18/20
Description
Describe the situation and context in detail
Min. 10 characters
0/100
Previous
Next

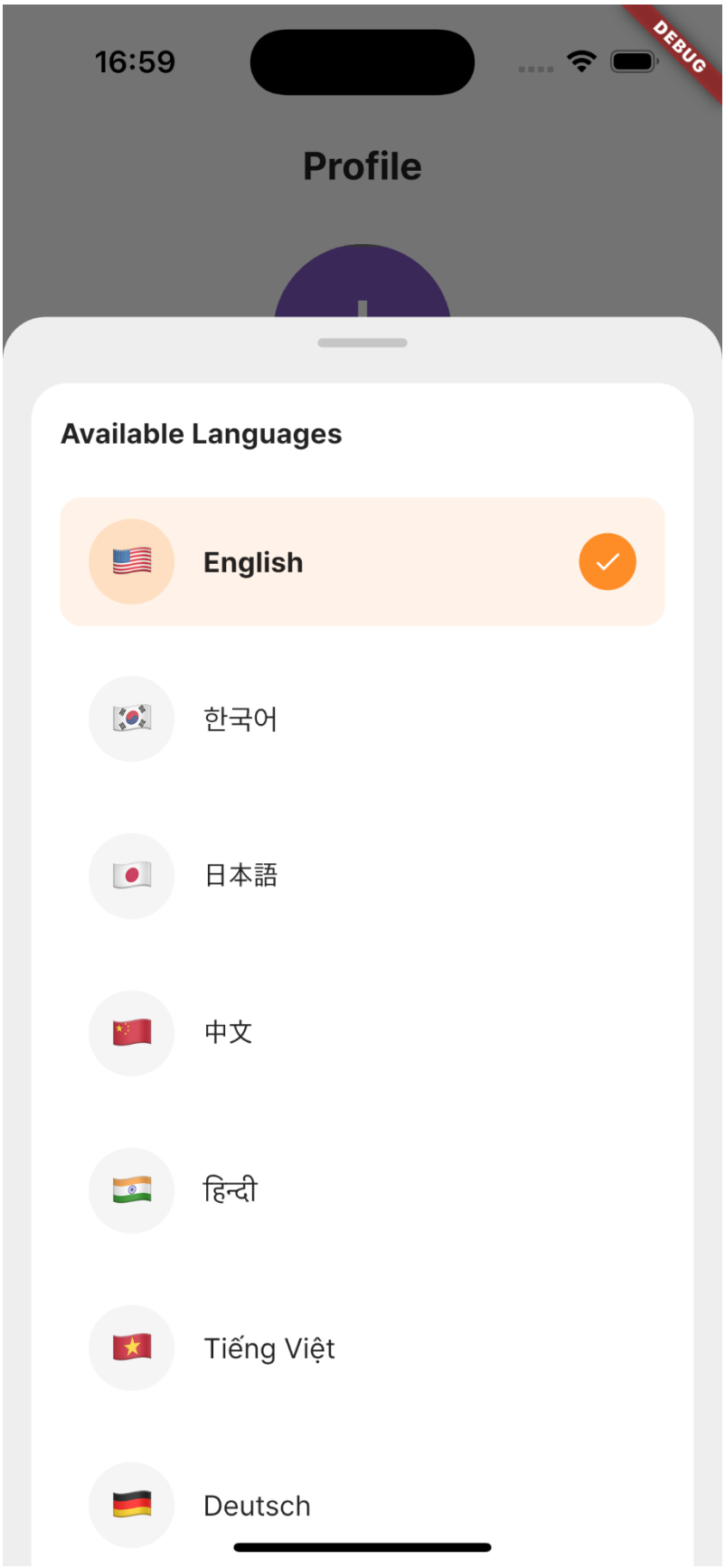

16:56
DEBUG
Create Custom Role Play
AI Voice Selection
Choose a voice that matches your AI character
Auto
Automatic voice generation based on...
Friendly
Warm and kind voice
Calm
Quiet voice with stability
Dramatic
Expressive emotional voice
Serene
Professional
Previous
Next


16:57
DEBUG
Create Custom Role Play
Role Play Visibility
Public Role Play
Other users can see this role play
Previous
Create


16:57
DEBUG
Create Custom Role Play
1/5
Collecting ideas...
AI is creating your role play. Please wait a moment. It takes about 30 seconds on average!



Language Settings:


16:59
DEBUG
Profile
Available Languages
English
한국어
日本語
中文
हिन्दी
Tiếng Việt
Deutsch

XI. Appendix B: User Testing Questionnaires

Conversation App - Usability Testing

Form description

What is your age? *

- Under 18
- 18 - 24
- 24 -34
- 35 - 44
- 45 +

Are you a domestic or international student? *

- Domestic student
- International student

What is your native language? *

Short answer text

How long have you been living in your current country of study? *

Short answer text

Section 2 of 7

App Interface

Description (optional)

How easy was it to navigate through the app upon first use? *

	1	2	3	4	5	
Very Difficult	○	○	○	○	○	Very Easy

How visually appealing did you find the app's interface? *

	1	2	3	4	5	
Very Unappealing	○	○	○	○	○	Very Appealing

Were the features (situational roleplay, voice command, translation, custom situation) easy to find and use? *

- ○ Yes
- ○ No
- ○ Somewhat

If you find the features *not/somewhat* easy to find and use, please specify what was unclear

Long answer text

Section 3 of 7

Main Feature: Situational Roleplay

Description (optional)

Did the AI-generated roleplay situations feel realistic and relevant to your daily life as an international student? *

	1	2	3	4	5	
Never	○	○	○	○	○	Always

How natural did the AI responses feel during conversations? *

	1	2	3	4	5	
Very Artificial	○	○	○	○	○	Very Natural

Did you find the voice command feature accurate and responsive? *

- ○ Yes
- ○ No

Describe any issues you faced with the voice command feature (if any)

Long answer text

Section 4 of 7

Feedback and Learning Screen

Description (optional)

How useful did you find the feedback on your responses (e.g., grammar correction, improvement suggestions)? *

	1	2	3	4	5	
Not useful at all	○	○	○	○	○	Extremely useful

Did the language translation and pronunciation feature help you understand and learn better? *

- ○ Yes
- ○ No
- ○ A little

Section 5 of 7

Custom Scenario

Description (optional)

How easy was it to create a custom roleplay situation? *

	1	2	3	4	5	
Very Difficult	○	○	○	○	○	Very Easy

Section 6 of 7

Overall Satisfaction

Description (optional)

How confident do you feel making conversation with people after using the app? *

	1	2	3	4	5	
Much less confident	○	○	○	○	○	Extremely confident

Would you recommend this app to other students? *

	1	2	3	4	5	
Definitely Not	○	○	○	○	○	Definitely

Section 7 of 7

Suggestions and Improvements

Description (optional)

What was your favorite feature of the app, and why? *

Long answer text

What feature(s) did you find least useful or most confusing? *

Long answer text

Would you find a Progress Tracker feature beneficial? *

	1	2	3	4	5	
Not beneficial at all	○	○	○	○	○	Very beneficial

What improvements or new features would you suggest for the app?

Long answer text

Any additional comments or feedback:

Long answer text